Main points of the page: *Embedding to non-Euclidean Spaces* in UMAP

This page explains that **UMAP is not limited to embedding data into ordinary Euclidean (flat) space**. By changing the `output_metric`, UMAP can embed data onto other geometric spaces that may better reflect the structure of the data. [Embedding...umentation]

1. Standard (Plane) Embeddings

- UMAP normally embeds data into a 2D Euclidean plane.
- This is the familiar scatterplot visualization most users work with. [Embedding...umentation]

2. Spherical Embeddings

- UMAP can embed data onto the surface of a sphere using the **Haversine metric**.
- This can be useful when data has periodic or wrap-around properties where no point should be infinitely far from another.
- The resulting coordinates need to be transformed into 3D spherical coordinates for proper visualization. [Embedding...umentation]

3. Custom Output Spaces

- Users can provide a custom `output_metric`.
- The metric must return:
    - a distance value, and
    - the gradient of that distance (needed for optimization).
- The page demonstrates this with a **torus (donut-shaped surface)** by implementing a custom distance function and gradient. [Embedding...umentation]

4. Torus Embeddings

- A torus is represented using coordinates that wrap around in both directions.
- Distances account for the shorter path, including wrap-around movement.
- UMAP successfully embeds data onto the torus surface and the result can be visualized in 3D. [Embedding...umentation]

5. Practical Example: Embedding into Gaussian Distributions

Instead of embedding each point as a single location:

- Each sample is embedded as a **2D Gaussian distribution**.
- The embedding consists of:
    - 2 coordinates for the Gaussian mean,
    - 3 parameters describing covariance (size, shape, orientation).
- This allows UMAP to represent **uncertainty** in the embedding.

- Points between clusters often receive larger covariance, indicating greater uncertainty.
- Density plots can then be constructed from the collection of Gaussian distributions. [Embedding...umentation]

6. Hyperbolic Embeddings

The page concludes with a hyperbolic geometry example.

- Hyperbolic spaces are useful for data with hierarchical or tree-like structure.
- Direct optimization in the **Poincaré disk** is difficult because it has a hard boundary.
- UMAP instead uses the **hyperboloid model**, which is easier to optimize.
- After embedding in hyperboloid coordinates, the result can be transformed into a Poincaré disk visualization. [Embedding...umentation]

Key takeaway

The central message is that **UMAP's output space is highly flexible**. By selecting or defining appropriate output metrics, you can embed data into:

- Euclidean planes,
- spheres,
- tori,
- spaces of Gaussian distributions,
- hyperbolic manifolds,

and potentially other custom geometric spaces, provided you can define a suitable distance function and gradient. [Embedding...umentation]

Given your interest in hyperbolic UMAP, the most relevant section is the final one: it recommends using the **hyperboloid parameterization internally and converting to a Poincaré disk for visualization**, because optimization is much more stable in the hyperboloid model. [Embedding...umentation]

On the API page, UMAP lists the following valid string metrics for the `metric=` parameter, which controls how distances are measured in the **input space** before the embedding is built. [UMAP API G...umentation]

Continuous-valued metrics

Metric	What it measures
`euclidean`	Straight-line (L2) distance. Most common default.

`manhattan`	Sum of absolute coordinate differences (L1 distance).
`chebyshev`	Maximum coordinate difference.
`minkowski`	Generalization of L1 and L2; controlled by parameter `p`.
`canberra`	Relative difference, emphasizing small values.
`braycurtis`	Ratio-based dissimilarity often used in ecology.
`mahalanobis`	Distance accounting for covariance structure.
`wminkowski`	Weighted Minkowski distance.
`seuclidean`	Standardized Euclidean distance (variance-scaled).
`cosine`	Angle between vectors, ignoring magnitude.
`correlation`	Similarity based on correlation rather than magnitude.
`haversine`	Great-circle distance on a sphere, useful for latitude/longitude data. [UMAP API G...umentation]

Binary / set-based metrics

These are often useful for sparse vectors, fingerprints, presence/absence data, bag-of-words representations, etc. [UMAP API G...umentation]

Metric	Idea	
`hamming`	Fraction of positions that differ.	
`jaccard`	Difference between sets based on intersection vs union.	
`dice`	Similar to Jaccard but weights shared positives differently.	

`russellrao`	Similarity based on simultaneous presence of features.	
`kulsinski`	Binary mismatch metric.	
`rogerstanimoto`	Binary agreement/disagreement measure.	
`sokalmichener`	Symmetric binary similarity measure.	
`sokalsneath`	Binary similarity emphasizing mismatches.	
`yule`	Based on odds-ratio relationships in binary features.	[UMAP API G...umentation]

Probability-distribution metrics

Metric	Use case
`hellinger`	Distance between probability distributions.
`ll_dirichlet`	Dirichlet-based likelihood metric for compositional/probability data. [UMAP API G...umentation]

Which metrics are commonly useful?

For typical machine-learning embeddings:

- **Dense numerical features** → `euclidean`, `manhattan`, `mahalanobis`
- **Embeddings from LLMs or transformers** → `cosine` is often the best starting point
- **Text bag-of-words / sparse vectors** → `cosine`, `jaccard`, or `dice`
- **Probability vectors** → `hellinger`
- **Geographic coordinates** → `haversine` [UMAP API G...umentation]

Given your open tabs about **hyperbolic embeddings**, **hyperspheres**, and **embedding projectors**, I'd particularly highlight:

- `cosine` because it approximates angular geometry and is widely used for semantic embeddings.
- `haversine` because it's one of the few explicitly non-Euclidean metrics built into UMAP.

- `mahalanobis` because it introduces local anisotropic geometry through a covariance matrix and can sometimes reveal structure missed by Euclidean distance. [UMAP API G...umentation]

Interestingly, **hyperbolic distance is not listed as an input metric here**. Instead, UMAP's hyperbolic support is exposed through the **`output_metric='hyperboloid'`** embedding space described in the non-Euclidean tutorial rather than as one of the standard input-space metrics. [Embedding...umentation], [UMAP API G...umentation]

Think of UMAP as having **two geometries**:

1. **Input geometry**: how distances are measured in your original data.
2. **Output geometry**: what kind of space UMAP is allowed to draw the embedding in.

The API exposes these as `metric` and `output_metric`. [UMAP API G...umentation], [Embedding...umentation]

Input metric (`metric=`)

The input metric tells UMAP:

> "When deciding which points are neighbors, how should distance be measured in the original high-dimensional dataset?"

For example:

umap.UMAP(metric="euclidean")

UMAP builds its neighborhood graph using Euclidean distance. [UMAP API G...umentation]

Or:

umap.UMAP(metric="cosine")

Now two vectors that point in similar directions are considered close, even if their magnitudes differ greatly. [UMAP API G...umentation]

Example

Suppose your data are text embeddings:

A = [1, 1]
B = [10, 10]
C = [1, -1]

- Euclidean says A and B are far apart.
- Cosine says A and B are identical because they point in the same direction.

Changing the **input metric** changes the neighborhood graph that UMAP learns from. [UMAP API G...umentation]

Output metric (`output_metric=`)

The output metric tells UMAP:

> "What kind of low-dimensional space should the embedding live in?"

Normally:

```
umap.UMAP(
metric="cosine",
output_metric="euclidean"
)
```

UMAP learns from cosine similarity but places points into an ordinary 2D plane. [UMAP API G...umentation], [Embedding...umentation]

You can instead choose:

```
umap.UMAP(
metric="cosine",
output_metric="haversine"
)
```

Now the optimization occurs on a spherical manifold rather than a flat plane. [Embedding...umentation]

Or:

```
umap.UMAP(
metric="cosine",
```

output_metric="hyperboloid"
)

Now the embedding lives in hyperbolic space. [Embedding...umentation]

A useful mental picture

Imagine UMAP as a cartographer.

Step 1: Measure the world

Input metric:

metric = ?

Determines which cities are near each other.

A ---- B
|
|
C

Step 2: Draw the map

Output metric:

output_metric = ?

Determines whether the map is drawn on:

- a sheet of paper (Euclidean)
- a globe (sphere)
- a donut (torus)
- a hyperbolic surface (hyperboloid) [Embedding...umentation]

Why hyperbolic output is interesting

Hyperbolic space expands exponentially as you move away from the center.

That means:

Tree structure

Root
/
A B
/ | \ / |
...

can often be represented more naturally than in Euclidean space. Hierarchies, taxonomies, graphs, and many language datasets have this property. [Embedding...umentation]

You might therefore use:

umap.UMAP(
metric="cosine",
output_metric="hyperboloid"
)

which means:

- Neighbors are determined using cosine similarity.
- The final embedding is optimized in hyperbolic geometry. [Embedding...umentation], [UMAP API G...umentation]

An important subtlety

The two metrics do **not** need to be related.

For example:

umap.UMAP(
metric="jaccard",
output_metric="hyperboloid"
)

is perfectly valid.

Interpretation:

- Original data are compared as sets via Jaccard distance.
- The resulting manifold is embedded into hyperbolic space. [UMAP API G...umentation], [Embedding...umentation]

For your hypersphere/hyperbolic exploration

A useful way to think about it is:

Question	Parameter
How should similarity be computed in the source data?	`metric`
What geometry should the embedding itself live in?	`output_metric`

If you're working with modern embedding vectors (transformers, LLM embeddings, sentence embeddings), a common advanced combination is:

umap.UMAP(
metric="cosine",
output_metric="hyperboloid"
)

because cosine often matches semantic similarity well, while hyperbolic space can better accommodate hierarchical structure in the resulting embedding. [UMAP API G...umentation], [Embedding...umentation]

Here's a complete example showing the difference between **input metric** and **output metric**.

The example uses:

- **Input metric:** `cosine`
- **Output metric:** `hyperboloid` (hyperbolic space)

This means:

1. UMAP finds neighbors using cosine similarity in the original high-dimensional data.
2. UMAP optimizes the embedding in hyperbolic geometry using the hyperboloid model. [UMAP API G...umentation], [Embedding...umentation]

```
import numpy as np
import matplotlib.pyplot as plt
import umap
from sklearn.datasets import load_digits

# Load example data
digits = load_digits()
X = digits.data
y = digits.target

# Hyperbolic UMAP
mapper = umap.UMAP(
n_neighbors=15,
min_dist=0.05,
metric="cosine", # input metric
output_metric="hyperboloid", # output metric
random_state=42
)

embedding = mapper.fit_transform(X)

# Hyperboloid coordinates
x = embedding[:, 0]
y_coord = embedding[:, 1]

# Hyperboloid z-coordinate
z = np.sqrt(1 + np.sum(embedding**2, axis=1))

# Convert to Poincare disk
disk_x = x / (1 + z)
disk_y = y_coord / (1 + z)

# Plot
fig, ax = plt.subplots(figsize=(8, 8))

scatter = ax.scatter(
disk_x,
```

```
disk_y,
c=digits.target,
cmap="Spectral",
s=20
)

# Draw boundary circle
boundary = plt.Circle(
(0, 0),
1.0,
fill=False,
color="black",
linewidth=2
)
ax.add_artist(boundary)

ax.set_aspect("equal")
ax.set_title("Cosine Input Metric + Hyperbolic Output Metric")
ax.axis("off")

plt.show()
```

This follows the hyperbolic embedding workflow described in the UMAP non-Euclidean embedding tutorial, which embeds in the hyperboloid model and then projects the result to the Poincaré disk for visualization. [Embedding...umentation]

Example 2: Same input metric, Euclidean output

```
mapper = umap.UMAP(
metric="cosine",
output_metric="euclidean",
random_state=42
)

embedding = mapper.fit_transform(X)

plt.scatter(
```

```
embedding[:,0],
embedding[:,1],
c=digits.target,
cmap="Spectral",
s=20
)
plt.title("Cosine Input Metric + Euclidean Output Metric")
plt.show()
```

Here the neighborhood graph is identical to the previous example because the input metric is still cosine. The difference is that the embedding is forced into a flat Euclidean plane instead of hyperbolic space. [UMAP API G...umentation], [Embedding...umentation]

Example 3: Spherical output

```
mapper = umap.UMAP(
metric="cosine",
output_metric="haversine",
random_state=42
)

embedding = mapper.fit_transform(X)

lat = embedding[:,0]
lon = embedding[:,1]

x = np.sin(lat) * np.cos(lon)
y = np.sin(lat) * np.sin(lon)
z = np.cos(lat)
```

In this case:

- neighbors are computed using cosine similarity,
- optimization occurs on a sphere using the Haversine distance metric,
- the resulting coordinates can be visualized on a globe. [Embedding...umentation], [UMAP API G...umentation]

For transformer embeddings

A practical configuration is:

```
mapper = umap.UMAP(
n_neighbors=50,
min_dist=0.0,
metric="cosine",
output_metric="hyperboloid",
random_state=42
)
```

Because transformer vectors are often compared with cosine similarity, while hierarchical semantic structure may be represented more naturally in hyperbolic space. The UMAP documentation specifically supports hyperbolic embeddings through `output_metric="hyperboloid"` rather than through the input metric list. [UMAP API G...umentation], [Embedding...umentation]

You can extend the previous example to show **both**:

1. The embedding on the **hyperboloid surface** (3D).
2. The corresponding **Poincaré disk projection** (2D).

The UMAP documentation's hyperbolic example computes the third coordinate as

```
z = np.sqrt(1 + np.sum(embedding**2, axis=1))
```

and then projects to the Poincaré disk using

```
disk_x = x / (1 + z)
disk_y = y / (1 + z)
```

after embedding with `output_metric="hyperboloid"`. [Embedding...umentation], [UMAP API G...umentation]

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import load_digits
import umap

# ----------------------------
```

```
# Data
# ----------------------------
digits = load_digits()

X = digits.data
labels = digits.target

# ----------------------------
# Hyperbolic UMAP
# ----------------------------
mapper = umap.UMAP(
n_neighbors=15,
min_dist=0.05,
metric="cosine", # input metric
output_metric="hyperboloid", # output geometry
random_state=42
)

embedding = mapper.fit_transform(X)

# ----------------------------
# Hyperboloid coordinates
# ----------------------------
x = embedding[:, 0]
y = embedding[:, 1]

# Hyperboloid model:
# z^2 - x^2 - y^2 = 1
z = np.sqrt(1 + x**2 + y**2)

# ----------------------------
# Poincare disk projection
# ----------------------------
disk_x = x / (1 + z)
disk_y = y / (1 + z)

# ----------------------------
```

```
# Figure
# ----------------------------
fig = plt.figure(figsize=(16, 7))

# ==================================
# LEFT: Hyperboloid in 3D
# ==================================
ax1 = fig.add_subplot(121, projection="3d")

scatter = ax1.scatter(
x,
y,
z,
c=labels,
cmap="Spectral",
s=20
)

ax1.set_title("Hyperboloid Embedding (3D)")
ax1.set_xlabel("x")
ax1.set_ylabel("y")
ax1.set_zlabel("z")

ax1.view_init(elev=30, azim=60)

# ==================================
# RIGHT: Poincare Disk
# ==================================
ax2 = fig.add_subplot(122)

ax2.scatter(
disk_x,
disk_y,
c=labels,
cmap="Spectral",
s=20
)
```

```
boundary = plt.Circle(
(0, 0),
1.0,
fill=False,
color="black",
linewidth=2
)

ax2.add_artist(boundary)

ax2.set_aspect("equal")
ax2.set_title("Poincaré Disk Projection")
ax2.axis("off")

plt.tight_layout()
plt.show()
```

What you're seeing

The 3D plot shows points on the **upper sheet of a two-sheet hyperboloid**:

z2−x2−y2=1z^2 - x^2 - y^2 = 1

This is the model UMAP actually optimizes in for `output_metric="hyperboloid"`. [Embedding...umentation]

The Poincaré disk is just a visualization of the same geometry. Points near the edge of the disk are not merely "farther away" in the Euclidean sense; the boundary represents points approaching infinite hyperbolic distance. The UMAP tutorial specifically projects the hyperboloid embedding into the disk after optimization. [Embedding...umentation]

For an even better visualization, you can also draw the hyperboloid surface mesh behind the embedded points:

```
u = np.linspace(0, 2*np.pi, 100)
v = np.linspace(0, 2.5, 60)

U, V = np.meshgrid(u, v)
```

```
Xsurf = np.sinh(V) * np.cos(U)
Ysurf = np.sinh(V) * np.sin(U)
Zsurf = np.cosh(V)

ax1.plot_surface(
Xsurf,
Ysurf,
Zsurf,
alpha=0.15,
color="lightgray",
linewidth=0
)
```

That makes it visually obvious that the embedding lives **on a curved manifold rather than in ordinary 3D Euclidean space**. [Embedding...umentation]

Here's a version that adds a **semi-transparent hyperboloid surface mesh** behind the UMAP points, making it much easier to see that the embedding lies on the hyperboloid model of hyperbolic space. The hyperboloid model used by UMAP satisfies
z2−x2−y2=1z^2 - x^2 - y^2 = 1

and the UMAP tutorial computes the hyperboloid coordinates and then projects them to the Poincaré disk for visualization. [Embedding...umentation], [UMAP API G...umentation]

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import load_digits
import umap

# ----------------------------
# Load data
# ----------------------------
digits = load_digits()
X = digits.data
labels = digits.target

# ----------------------------
# Hyperbolic UMAP
```

```
# ----------------------------
mapper = umap.UMAP(
n_neighbors=15,
min_dist=0.05,
metric="cosine",
output_metric="hyperboloid",
random_state=42
)

embedding = mapper.fit_transform(X)

# ----------------------------
# Hyperboloid coordinates
# ----------------------------
x = embedding[:, 0]
y = embedding[:, 1]
z = np.sqrt(1 + x**2 + y**2)

# ----------------------------
# Poincare disk projection
# ----------------------------
disk_x = x / (1 + z)
disk_y = y / (1 + z)

# ----------------------------
# Hyperboloid surface mesh
# ----------------------------
theta = np.linspace(0, 2 * np.pi, 120)
radius = np.linspace(0, 3.0, 80)

Theta, Radius = np.meshgrid(theta, radius)

Xsurf = np.sinh(Radius) * np.cos(Theta)
Ysurf = np.sinh(Radius) * np.sin(Theta)
Zsurf = np.cosh(Radius)

# ----------------------------
```

```
# Figure
# ----------------------------
fig = plt.figure(figsize=(18, 8))

# ====================================
# Left: Hyperboloid Model
# ====================================
ax1 = fig.add_subplot(121, projection="3d")

# surface mesh
ax1.plot_surface(
Xsurf,
Ysurf,
Zsurf,
color="lightgray",
alpha=0.25,
edgecolor="none",
zorder=0
)

# wireframe (optional but very helpful)
ax1.plot_wireframe(
Xsurf,
Ysurf,
Zsurf,
color="gray",
linewidth=0.2,
alpha=0.25
)

# data points
scatter = ax1.scatter(
x,
y,
z,
c=labels,
cmap="Spectral",
```

```
s=25,
depthshade=True
)

ax1.set_title("Hyperboloid Model")
ax1.set_xlabel("x")
ax1.set_ylabel("y")
ax1.set_zlabel("z")

ax1.view_init(elev=28, azim=60)

# ====================================
# Right: Poincaré Disk
# ====================================
ax2 = fig.add_subplot(122)

ax2.scatter(
disk_x,
disk_y,
c=labels,
cmap="Spectral",
s=25
)

boundary = plt.Circle(
(0, 0),
1.0,
fill=False,
color="black",
linewidth=2
)

ax2.add_artist(boundary)

ax2.set_title("Poincaré Disk Projection")
ax2.set_aspect("equal")
ax2.axis("off")
```

plt.tight_layout()
plt.show()

Optional enhancements

You can make the geometry even more obvious:

ax1.set_box_aspect([1, 1, 1.5])

to vertically stretch the hyperboloid.

Or color the surface by hyperbolic radius:

colors = plt.cm.viridis(Radius / Radius.max())

ax1.plot_surface(
Xsurf,
Ysurf,
Zsurf,
facecolors=colors,
alpha=0.3,
linewidth=0
)

This creates a very intuitive side-by-side view:

- **Left:** the actual hyperboloid manifold where UMAP performs optimization.
- **Right:** the same points projected into the Poincaré disk, where the outer circle represents points approaching infinite hyperbolic distance. [Embedding...umentation]

You can animate the rotation of the hyperboloid using `matplotlib.animation.FuncAnimation` and save it as a GIF or MP4.

import numpy as np
import matplotlib.pyplot as plt
from matplotlib.animation import FuncAnimation
from sklearn.datasets import load_digits
import umap

```
# Data
# ----------------------------
digits = load_digits()

mapper = umap.UMAP(
metric="cosine",
output_metric="hyperboloid",
random_state=42
)

embedding = mapper.fit_transform(digits.data)

# Hyperboloid coordinates
x = embedding[:, 0]
y = embedding[:, 1]
z = np.sqrt(1 + x**2 + y**2)

# ----------------------------
# Hyperboloid mesh
# ----------------------------
theta = np.linspace(0, 2*np.pi, 120)
radius = np.linspace(0, 3.0, 80)

Theta, Radius = np.meshgrid(theta, radius)

Xsurf = np.sinh(Radius) * np.cos(Theta)
Ysurf = np.sinh(Radius) * np.sin(Theta)
Zsurf = np.cosh(Radius)

# ----------------------------
# Figure
# ----------------------------
fig = plt.figure(figsize=(10, 8))
ax = fig.add_subplot(111, projection="3d")

# Surface
ax.plot_surface(
```

```
Xsurf,
Ysurf,
Zsurf,
color="lightgray",
alpha=0.25,
linewidth=0,
)

# Wireframe
ax.plot_wireframe(
Xsurf,
Ysurf,
Zsurf,
color="gray",
linewidth=0.2,
alpha=0.3
)

# Embedded points
scatter = ax.scatter(
x, y, z,
c=digits.target,
cmap="Spectral",
s=25
)

ax.set_title("Hyperbolic UMAP on Hyperboloid")
ax.set_box_aspect([1, 1, 1.5])

# ----------------------------
# Animation
# ----------------------------
def update(frame):
ax.view_init(
elev=25 + 10np.sin(frame*np.pi/180),
azim=frame
)
```

```
return [scatter]

anim = FuncAnimation(
fig,
update,
frames=np.arange(0, 360, 2),
interval=50,
blit=False
)

# Save GIF
anim.save(
"hyperboloid_rotation.gif",
writer="pillow",
fps=20
)

# Or save MP4
# anim.save(
# "hyperboloid_rotation.mp4",
# writer="ffmpeg",
# fps=30
# )

plt.show()
```

More interesting animation

Instead of simply rotating, you can animate a geodesic-style fly-through:

```
def update(frame):

azim = frame
elev = 20 + 15*np.sin(np.radians(frame))

*ax.view_init(*
*elev=elev,*
*azim=azim*
```

)

dist = 8 + 2np.sin(np.radians(frame))
ax.dist = dist

return []

For Jupyter

Display directly in a notebook:

from IPython.display import HTML

HTML(anim.to_jshtml())

or

HTML(anim.to_html5_video())

Nice enhancement

Color the hyperboloid surface by hyperbolic radius:

colors = plt.cm.plasma(
Radius / Radius.max()
)

ax.plot_surface(
Xsurf,
Ysurf,
Zsurf,
facecolors=colors,
alpha=0.35,
shade=False
)

The resulting animation makes it much easier to see that the embedding points lie on the curved hyperboloid manifold and how that manifold corresponds to the Poincaré disk projection.

For a high-quality MP4 suitable for presentations, use:

- **1920×1080 (1080p)** or **3840×2160 (4K)** output
- **H.264 encoding**
- **30-60 FPS**
- Higher DPI for crisp text and points

Here's a production-quality export:

```
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.animation import FuncAnimation, FFMpegWriter
from sklearn.datasets import load_digits
import umap

# ----------------------------
# Data
# ----------------------------
digits = load_digits()

mapper = umap.UMAP(
metric="cosine",
output_metric="hyperboloid",
random_state=42
)

embedding = mapper.fit_transform(digits.data)

# Hyperboloid coordinates
x = embedding[:, 0]
y = embedding[:, 1]
z = np.sqrt(1 + x**2 + y**2)

# ----------------------------
# Hyperboloid mesh
# ----------------------------
theta = np.linspace(0, 2*np.pi, 180)
radius = np.linspace(0, 3.2, 120)

Theta, Radius = np.meshgrid(theta, radius)
```

```
Xsurf = np.sinh(Radius) * np.cos(Theta)
Ysurf = np.sinh(Radius) * np.sin(Theta)
Zsurf = np.cosh(Radius)

# ----------------------------
# Figure
# ----------------------------
fig = plt.figure(
figsize=(16, 9),
dpi=200
)

ax = fig.add_subplot(
111,
projection="3d"
)

# Colored surface
colors = plt.cm.plasma(
Radius / Radius.max()
)

ax.plot_surface(
Xsurf,
Ysurf,
Zsurf,
facecolors=colors,
linewidth=0,
alpha=0.35,
shade=False
)

# Wireframe
ax.plot_wireframe(
Xsurf,
Ysurf,
Zsurf,
```

```
color="black",
linewidth=0.15,
alpha=0.15
)

scatter = ax.scatter(
x,
y,
z,
c=digits.target,
cmap="Spectral",
s=20,
depthshade=True
)

ax.set_title(
"Hyperbolic UMAP Embedding",
fontsize=22,
pad=20
)

ax.set_box_aspect([1, 1, 1.4])

ax.set_xlabel("x")
ax.set_ylabel("y")
ax.set_zlabel("z")

# cleaner look
ax.grid(False)

# ------------------------------
# Animation
# ------------------------------
def update(frame):

azimuth = frame
```

```
elevation = (
25
+ 10np.sin(
np.radians(frame)
)
)

ax.view_init(
elev=elevation,
azim=azimuth
)

return []

frames = np.arange(0, 360, 1)

anim = FuncAnimation(
fig,
update,
frames=frames,
interval=33,
blit=False
)

# ----------------------------
# High Resolution MP4
# ----------------------------
writer = FFMpegWriter(
fps=30,
bitrate=20000,
codec="h264"
)

anim.save(
"hyperboloid_umap_1080p.mp4",
writer=writer,
dpi=200
```

)

4K export

For true 4K:

fig = plt.figure(
figsize=(19.2, 10.8),
dpi=200
)

This renders approximately:

3840 × 2160

and export with:

writer = FFMpegWriter(
fps=60,
bitrate=40000,
codec="h264"
)

Professional-quality settings

fps=60
bitrate=40000
dpi=200

These settings typically produce a very smooth rotating hyperboloid suitable for Keynote, PowerPoint, conference talks, or large displays.

Alternative cinematic camera path

Rather than a simple spin:

def update(frame):

t = np.radians(frame)

azim = frame

```
elev = 20 + 15*np.sin(t)
dist = 8 + np.sin(2*t)

ax.view_init(
elev=elev,
azim=azim
)

ax.dist = dist

return []
```

This creates a subtle orbit-and-zoom effect that gives a much stronger 3D impression of the hyperboloid surface.

If your goal is not just to generate a hyperbolic UMAP animation, but to build a system that takes someone from **zero mathematical background** to being able to **visualize, reason about, measure distances, compute angles, and navigate spaces of arbitrary dimensionality**, then I'd expand the project specification substantially.

Expanded Learning Objective

The animation becomes one module in a larger progression:

> Learn how to represent, visualize, transform, compare, and measure points embedded in spaces of dimensionality nn, where nn may be 2, 3, 10, 384, 768, or higher.

The learner should end with the ability to:

- Understand coordinates
- Understand vectors
- Compute distances
- Compute angles
- Understand projections
- Understand embeddings
- Understand manifolds
- Understand curvature
- Understand hyperbolic geometry
- Understand spherical geometry

- Understand dimensionality reduction
- Understand UMAP's input and output metrics
- Understand how high-dimensional semantic embeddings are represented geometrically

---

Suggested Curriculum

Level 0: Points

Start with:

Number line

A point:

x = 3

Distance:

|a-b|

Concept:

> Coordinates are addresses.

Level 1: 2D Geometry

Points:

A = (2,3)
B = (5,7)

Distance:

d=(x_2−x_1)2+(y_2−y_1)2d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}

Angle:

θ=arctan(y/x)\theta=\arctan(y/x)

Visualize:

- Cartesian plane
- Circle
- Unit circle

Level 2: 3D Geometry

Points:

(x,y,z)

Distance:

d=Δx2+Δy2+Δz2d=\sqrt{\Delta x^2+\Delta y^2+\Delta z^2}

Introduce:

- rotations
- camera transforms
- projections

This connects directly to your rotating hyperboloid animation.

Level 3: Vectors

Introduce:

v = [1,2,3]

Norm:

∣∣v∣∣||v||

Dot Product:

u⋅vu \cdot v

Angle:

cos(θ)=u·v||u||||v||\cos(\theta)= \frac{u\cdot v} {||u||||v||}

This is the foundation for:

metric="cosine"

in UMAP. [UMAP API G...umentation]

Level 4: N-Dimensional Spaces

Generalize:

[x1,x2,x3,...,xn]

Distance:

d=∑_i=1n(x_i−y_i)2d= \sqrt{ \sum_{i=1}^{n} (x_i-y_i)^2 }

Nothing fundamentally changes.

A 768-dimensional embedding is just:

[0.17,
-0.82,
...
0.33]

with 768 coordinates.

Level 5: Metric Spaces

Introduce:

Distance need not be Euclidean.

Examples:

euclidean
manhattan
cosine
jaccard
haversine

UMAP uses these as input metrics. [UMAP API G...umentation]

Concept:

> Geometry is determined by the distance function.

Level 6: Inner Products

Introduce:

⟨x,y⟩\langle x,y \rangle

This yields:

- angles
- lengths
- orthogonality

Important because many machine-learning embeddings are essentially geometric objects defined through inner products.

Level 7: Linear Algebra

Required concepts:

- vectors
- matrices
- matrix multiplication
- eigenvectors
- eigenvalues

Understanding:

PCA
SVD
Spectral Embeddings

Level 8: Manifolds

Key intuition:

Earth appears flat locally:

R^2

but globally:

Sphere

A manifold is:

> A space that looks Euclidean when viewed locally.

UMAP assumes data lies on an underlying manifold. [UMAP API G...umentation]

Level 9: Curvature

Three geometries:

Euclidean

Flat
$K=0$

Spherical

Positive curvature
$K>0$

Hyperbolic

Negative curvature

K<0K<0

This is where the UMAP spherical and hyperboloid output metrics become meaningful. [Embedding...umentation]

Level 10: Hyperbolic Geometry

Understand:

Hyperboloid Model

z2−x2−y2=1z^2-x^2-y^2=1

Used internally by UMAP. [Embedding...umentation]

Poincaré Disk

Projection:

X=x1+zX=\frac{x}{1+z} Y=y1+zY=\frac{y}{1+z}

Also described in the UMAP hyperbolic tutorial. [Embedding...umentation]

Level 11: Geodesics

Rather than straight lines:

Measure along the manifold.

Examples:

- great circles on spheres
- hyperbolic geodesics on the disk

This is the true notion of shortest path.

Level 12: Embeddings

Distinguish:

Input Space

Original coordinates.

Example:

768 dimensions

Output Space

Embedding:

2 dimensions

or

hyperboloid

This directly corresponds to UMAP's `metric` and `output_metric`. [UMAP API G...umentation], [Embedding...umentation]

Level 13: Measuring Angles in N Dimensions

The exact same formula works:

cos(θ)============∑x_iy_i‖x‖‖y‖\cos(\theta) ============ \frac{ \sum x_i y_i }{ ||x|| ||y|| }

Whether:

2D

or

10000D

the mathematics is identical.

Level 14: Measuring Distances in N Dimensions

Euclidean:

∑(x_i−y_i)2\sqrt{ \sum (x_i-y_i)^2 }

Manhattan:

∑∣x_i−y_i∣\sum |x_i-y_i|

Cosine:

1−cos(θ)1-\cos(\theta)

Hyperbolic:

Uses the Lorentzian inner product.

Level 15: Visualization of Higher Dimensions

Methods:

Projection

N → 3

PCA

Linear compression.

UMAP

Nonlinear manifold-preserving compression. [UMAP API G...umentation]

t-SNE

Neighborhood-preserving compression.

Hyperbolic Projection

N → Hyperboloid

then

Hyperboloid → Poincaré Disk

[Embedding...umentation]

Final Capstone

Your final visualization system could simultaneously display:

Panel 1

Original feature space statistics

768 dimensions

Panel 2

3D Hyperboloid

Animated rotation

Panel 3

Poincaré Disk

Interactive geodesics

Panel 4

Distance inspector

Select two points.

Display:

Euclidean Distance
Cosine Distance
Angular Distance
Hyperbolic Distance
Jaccard Distance

Panel 5

Angle inspector

Display:

Dot Product
Angle
Orthogonality
Similarity

Panel 6

Dimension explorer

Move:

2D
3D
10D
100D
768D
1536D

and observe how distance distributions change.

That progression would take a learner from essentially no geometry background to being capable of understanding and working with the mathematical machinery behind high-dimensional embeddings, UMAP, spherical spaces, hyperbolic manifolds, and modern embedding models.

Beginner-Friendly Visual Glossary

A visual intuition guide for understanding high-dimensional spaces, embeddings, UMAP, and hyperbolic geometry

---

1. Point

Visual

•

What it is

A point is a location.

Think:

- A house on a map
- A city on a globe
- A row in a dataset

Example

Point A = (2,3)

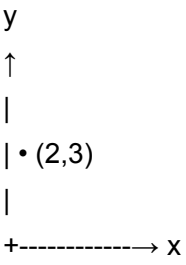
means:

- 2 units right
- 3 units up

---

2. Coordinate

Visual



What it is

A coordinate is an address that tells you where a point lives.

Examples:

1D: (5)

2D: (2,3)

3D: (2,3,4)

768D: $(x_1,x_2,x_3,\ldots,x_{768})$

---

3. Dimension

Visual

1D = line

2D = square

3D = cube

What it is

A dimension is a direction you can move independently.

Space	Dimensions
Number line	1
Sheet of paper	2
Physical room	3
Embedding vector	Hundreds or thousands

---

4. Vector

Visual

A ------>

What it is

A vector has:

- length
- direction

Example:

[1,2,3]

Vectors are the basic objects used in machine learning.

---

5. Distance

Visual

A •-------• B

What it is

Distance measures how far apart points are.

Example

A=(0,0)
B=(3,4)

Distance:

5

using:

d=x2+y2d=\sqrt{x^2+y^2}

---

6. Angle

Visual

B
/
/
/
O------A

What it is

Measures the difference in direction.

Common values

0° same direction

90° perpendicular

180° opposite

---

7. Dot Product

Visual

Similarity Detector

What it does

Determines:

- similarity
- angle
- alignment

Positive:

same direction

Zero:

perpendicular

Negative:

opposite direction

---

8. Norm (Length)

Visual

v ---------->

What it is

The length of a vector.

Example:

(3,4)

has norm:

5

---

9. Cosine Similarity

Visual

↗

↗ smaller angle = more similar

What it measures

Direction similarity instead of physical distance.

Useful for

- text embeddings
- semantic search
- LLM embeddings

Often the best input metric for embedding vectors.

---

10. Metric

Visual

Different rulers

What it is

A rule for measuring distance.

Examples:

Euclidean

Manhattan

Cosine

Jaccard

Hyperbolic

Different metrics create different geometries.

---

11. Euclidean Space

Visual

Flat paper

Geometry

□

△

○

Properties

- Flat
- Familiar
- Ordinary geometry

Curvature:

0

---

12. Sphere

Visual

.-' '-.

( )

'-..-'

Geometry

Points live on the surface.

Examples

- Earth
- Planets

Curvature

Positive.

---

13. Great Circle

Visual

Earth Equator

What it is

Shortest path on a sphere.

Example:

Aircraft routes.

---

14. Haversine Distance

Visual

🌎
A ---- B

What it measures

Distance along a sphere.

Useful for:

- latitude
- longitude
- planetary coordinates

---

15. Hyperplane

Visual

Line in 2D

Plane in 3D

Generalized plane in N dimensions

What it is

A flat surface inside a higher-dimensional space.

Important in:

- machine learning
- classification
- geometry

---

16. Manifold

Visual

🌎

What it is

A space that appears flat locally but curved globally.

Examples:

- sphere
- torus
- hyperboloid

UMAP assumes data lies on a manifold.

---

17. Curvature

Visual

Flat Curved

----)

Types

Zero

Euclidean

Positive

Sphere

Negative

Hyperbolic

---

18. Hyperbolic Space

Visual

Space expands rapidly

Intuition

Every step outward gives room for exponentially more points.

Ideal for:

- trees
- hierarchies
- taxonomies

---

19. Hyperboloid

Visual

/

/

/

\ /

\ /

Equation

z2−x2−y2=1z^2-x^2-y^2=1

What it is

A mathematical model of hyperbolic space.

Used directly by hyperbolic UMAP.

---

20. Poincaré Disk

Visual

_____

**/ **

| • |
_____/

What it is

A way of drawing infinite hyperbolic space inside a circle.

Key idea

The boundary means:

Infinite distance away

---

21. Geodesic

Visual

Shortest path on a curved surface

Examples

Sphere:

Great circle

Hyperbolic:

Curved arc

---

22. Embedding

Visual

High-D
↓
Low-D

What it is

A transformation that places complex objects into a geometric space.

Examples:

Words
Documents
Images
People
Graphs

may become points.

---

23. Dimensionality Reduction

Visual

768D
↓
2D

Goal

Keep important relationships while reducing complexity.

---

24. UMAP

Visual

High-D data
↓
Neighborhood graph
↓

Low-D embedding

Purpose

Creates a lower-dimensional representation while preserving structure.

---

25. Input Metric

Visual

How similarity is measured before embedding

Examples:

metric="cosine"
metric="euclidean"
metric="jaccard"

Question answered:

> Which points are neighbors?

---

26. Output Metric

Visual

What space the embedding lives in

Examples:

output_metric="euclidean"
output_metric="haversine"
output_metric="hyperboloid"

Question answered:

What geometry should UMAP draw in?

---

27. Projection

Visual

3D → 2D

Example

Shadow:

Object → Wall

A projection shows a higher-dimensional object using fewer dimensions.

---

28. PCA

Visual

Cloud
↓
Best-fitting plane

What it does

Compresses dimensions while preserving variance.

---

29. Semantic Embedding

Visual

Words → Points

Example:

Dog
Cat
Wolf

appear close together.

Dog
Galaxy

appear farther apart.

---

30. Hyper-Dimensional Space

Visual

2D
3D
10D
100D
768D
1536D

Key Insight

The formulas don't fundamentally change.

Distance in 768 dimensions is still:

d=∑_i=1768(x_i−y_i)2d= \sqrt{ \sum_{i=1}^{768} (x_i-y_i)^2 }

Angles are still computed from dot products.

The mathematics scales naturally to any number of dimensions.

One-Sentence Summary

A high-dimensional embedding is simply a collection of points with coordinates; metrics measure distances between them, angles measure directional similarity, manifolds describe the shape of the space they occupy, and tools like UMAP create lower-dimensional visualizations while trying to preserve those relationships.

Here's a single-file interactive HTML glossary that includes:

- ✅ Search box
- ✅ Expand/collapse cards
- ✅ Category filtering
- ✅ Visual ASCII diagrams
- ✅ Mathematical formulas
- ✅ Beginner-friendly explanations
- ✅ Responsive layout
- ✅ Easy future expansion

Save as `interactive_glossary.html` and open in any browser.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Geometry & Embedding Visual Glossary</title>

<style>

body{
font-family:Segoe UI,Arial,sans-serif;
background:#f4f6f9;
margin:0;
padding:30px;
color:#222;
}

h1{
```

```
text-align:center;
}

.subtitle{
text-align:center;
color:#555;
margin-bottom:30px;
}

.controls{
display:flex;
gap:10px;
flex-wrap:wrap;
justify-content:center;
margin-bottom:25px;
}

input{
padding:12px;
width:300px;
border-radius:8px;
border:1px solid #ccc;
}

button{
padding:10px 18px;
border:none;
border-radius:8px;
cursor:pointer;
background:#3366ff;
color:white;
}

button:hover{
background:#2248c9;
}
```

```
.card{
background:white;
border-radius:12px;
padding:18px;
margin:12px auto;
max-width:900px;
box-shadow:0 2px 10px rgba(0,0,0,.1);
}

.term{
font-size:22px;
font-weight:bold;
cursor:pointer;
}

.category{
color:#666;
font-size:12px;
text-transform:uppercase;
}

.content{
display:none;
margin-top:15px;
}

.visual{
background:#111;
color:#00ff90;
padding:15px;
border-radius:8px;
font-family:monospace;
overflow:auto;
white-space:pre;
}

.math{
```

```
background:#eef4ff;
padding:10px;
border-radius:8px;
margin:10px 0;
}

.show{
display:block;
}

</style>
</head>

<body>

<h1>Interactive Geometry & Embedding Glossary</h1>

<p class="subtitle">
From absolute beginner to high-dimensional geometry, manifolds,
UMAP, and hyperbolic space.
</p>

<div class="controls">

<input
id="searchBox"
placeholder="Search..."
onkeyup="filterTerms()"
>

<button onclick="expandAll()">Expand All</button>
<button onclick="collapseAll()">Collapse All</button>

</div>

<div id="glossary">
```

```
<!-- Point -->

<div class="card" data-name="point coordinate location">
<div class="category">Foundations</div>

<div class="term" onclick="toggle(this)">
Point
</div>

<div class="content">

<p><strong>A location.</strong></p>

<div class="visual">
•
</div>

<p>
A point is the most basic object in geometry.
</p>

</div>
</div>

<!-- Coordinate -->

<div class="card" data-name="coordinate address">
<div class="category">Foundations</div>

<div class="term" onclick="toggle(this)">
Coordinate
</div>

<div class="content">

<div class="visual">
y
```

↑
|
| • (2,3)
|
+----------→ x

A coordinate is an address that tells you where a point is located.

<!-- Dimension -->

Foundations

Dimension

1D = Line

2D = Plane

3D = Space

768D = Embedding Space

A dimension is an independent direction in which movement is possible.

```
</p>

</div>
</div>

<!-- Vector -->

<div class="card" data-name="vector direction length">
<div class="category">Linear Algebra</div>

<div class="term" onclick="toggle(this)">
Vector
</div>

<div class="content">

<div class="visual">
A -------->
</div>

<p>
A vector has both length and direction.
</p>

<div class="math">
Example:
[1, 2, 3]
</div>

</div>
</div>

<!-- Distance -->

<div class="card" data-name="distance euclidean">
<div class="category">Geometry</div>
```

Distance

A •-------• B

Distance measures how far apart two points are.

$d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$

<!-- Angle -->

Geometry

Angle

B
/
/
/

O------A

Measures difference in direction.

<!-- Dot Product -->

Linear Algebra

Dot Product

Measures vector alignment.

Positive = Similar

Zero = Perpendicular

Negative = Opposite

<!-- Cosine Similarity -->

Metrics

Cosine Similarity

↗
↗

Small angle
=
High similarity

Commonly used for embeddings and semantic search.

<!-- Metric -->

Metrics

Metric

Different rulers:

Euclidean
Manhattan
Cosine
Jaccard
Hyperbolic

A metric defines what "distance" means.

<!-- Manifold -->

Topology

Manifold

🌎

A space that looks flat locally but may be curved globally.

<!-- Curvature -->

Geometry

Curvature

Flat Spherical Hyperbolic

---- ○)

Zero = Euclidean

Positive = Sphere

Negative = Hyperbolic

<!-- Hyperbolic Space -->

Hyperbolic Geometry

Hyperbolic Space

Space expands
exponentially outward

Useful for hierarchies and tree-like data.

<!-- Hyperboloid -->

Hyperbolic Geometry

Hyperboloid

/
/
/
\ /
\ /

$z^2 - x^2 - y^2 = 1$

A model of hyperbolic space used by hyperbolic UMAP.

<!-- Poincare Disk -->

Hyperbolic Geometry

Poincaré Disk

_____

**/ **

| • |

______/

Represents infinite hyperbolic space inside a finite circle.

<!-- Embedding -->

Machine Learning

Embedding

Text / Images / Objects

↓

Vectors

↓

Geometry

Represents complex entities as points in a space.

<!-- UMAP -->

Machine Learning

UMAP

High-D Data

↓

Neighborhood Graph

↓

Low-D Embedding

A dimensionality reduction algorithm preserving local structure.

<!-- Input Metric -->

UMAP

Input Metric

How distance is measured before embedding.

metric="cosine"

metric="euclidean"

metric="jaccard"

<!-- Output Metric -->

UMAP

Output Metric

Defines the geometry of the embedding space.

output_metric="euclidean"

output_metric="haversine"

output_metric="hyperboloid"

```
</div>
</div>

</div>

<script>

function toggle(element)
{
const content =
element.nextElementSibling;

content.classList.toggle("show");
}

function expandAll()
{
document
.querySelectorAll('.content')
.forEach(c=>c.classList.add('show'));
}

function collapseAll()
{
document
.querySelectorAll('.content')
.forEach(c=>c.classList.remove('show'));
}

function filterTerms()
{
let query =
document
.getElementById('searchBox')
.value
.toLowerCase();
```

```
document
.querySelectorAll('.card')
.forEach(card =>
{
let text =
card.innerText.toLowerCase();

card.style.display =
text.includes(query)
? 'block'
: 'none';
});
}

</script>

</body>
</html>
```

Recommended next upgrades

1. **MathJax support** for beautifully rendered equations.
2. **Interactive 2D/3D diagrams** using Plotly.
3. **Rotating hyperboloid visualization** embedded directly in the glossary.
4. **Distance calculator** where users enter points in N dimensions.
5. **Angle calculator** with live cosine similarity.
6. **UMAP Playground** showing input metric vs output metric effects.
7. **Dimension Explorer** allowing users to move from 1D → 2D → 3D → 10D → 768D interactively.
8. **Animated Poincaré Disk and Hyperboloid models** with geodesics and shortest paths.

Those additions would turn the glossary into a full interactive learning environment for high-dimensional geometry and embeddings.

Add these sections to your HTML glossary. They introduce:

- **MathJax-rendered formulas**
- **Dimension selector (1D–1000D+)**
- **Euclidean distance calculator**
- **Manhattan distance calculator**
- **Cosine similarity calculator**

- **Angle calculator**
- Automatic vector generation for arbitrary dimensionality

---

1. Add MathJax to <head>

<script>
window.MathJax = {
tex: {
inlineMath: [['', '']],
displayMath: [['
', '
']]
}
};
</script>

<script
id="MathJax-script"
async
src="https://cdn.jsdelivr.net/npm/mathjax@3/es5/tex-mml-chtml.js">
</script>

Now formulas such as

[
d=\sqrt{\sum_{i=1}^{n}(x_i-y_i)^2}
]

render beautifully.

---

2. Add Advanced Mathematics Card

Place after the glossary entries.

```
<div class="category">
Interactive Mathematics Lab
</div>

<div class="term"
onclick="toggle(this)">
Distance & Angle Calculator
</div>

<div class="content">

<h3>Select Dimensionality</h3>

<input
type="number"
id="dimension"
value="3"
min="1"
max="1000"
>

<button onclick="generateVectors()">
Generate Inputs
</button>

<div id="vectorInputs"></div>

<button onclick="calculateMetrics()">
Compute
</button>

<div id="results"></div>

</div>
</div>
```

---

3. Add CSS

```
.vector-grid{
margin-top:15px;
}

.vector-grid input{
width:70px;
margin:3px;
}

.result-box{
margin-top:20px;
padding:15px;
border-radius:8px;
background:#eef4ff;
}
```

---

4. Add JavaScript

Place below your existing functions.

```
<script>

generateVectors();

function generateVectors()
{
let n =
parseInt(
document.getElementById(
"dimension"
).value
);

let html =
"<div class='vector-grid'>";
```

```
html += "<h4>Vector A</h4>";

for(let i=0;i<n;i++)
{
html +=
&lt;input</span><br></div><div><span
attribution="{&quot;name&quot;:&quot;Copilot&quot;,&quot;oid&quot;:&quot;E
64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E
12-4514-AD9B-893A6FAFD00C&quot;,&quot;userInfo&quot;:{&quot;name&quot;:&qu
ot;Copilot&quot;,&quot;oid&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00
C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;},&
quot;timestamp&quot;:1793397600000,&quot;dataSource&quot;:0}"> id="a
i" < /span >< br >< /div >< div >< spanattribution = "&quot;name&quot;:&quot;Copilot&quot;,&quot;oid&
{i+1}"&gt;</span><br></div><div><span
attribution="{&quot;name&quot;:&quot;Copilot&quot;,&quot;oid&quot;:&quot;E
64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E
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quot;timestamp&quot;:1793397600000,&quot;dataSource&quot;:0}"> ;
}

html += "<h4>Vector B</h4>";

for(let i=0;i<n;i++)
{
html +=
&lt;input</span><br></div><div><span
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```

```
quot;timestamp&quot;:1793397600000,&quot;dataSource&quot;:0}"> id="b
i" < /span >< br >< /div >< div >< spanattribution = "&quot; name&quot;: &quot; Copilot&quot;, &quot; oid&
{(i+1)*2}"&gt;</span><br></div><div><span
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quot;timestamp&quot;:1793397600000,&quot;dataSource&quot;:0}"> ;
}

html += "</div>";

document
.getElementById(
"vectorInputs"
).innerHTML = html;
}

function calculateMetrics()
{
let n =
parseInt(
document.getElementById(
"dimension"
).value
);

let A=[];
let B=[];

for(let i=0;i<n;i++)
{
A.push(
parseFloat(
document.getElementById(
```

```
a${i}
).value
)
);

B.push(
parseFloat(
document.getElementById(
b${i}
).value
)
);
}

let euclidean=0;
let manhattan=0;

let dot=0;

let normA=0;
let normB=0;

for(let i=0;i<n;i++)
{
let diff=A[i]-B[i];

euclidean += diff*diff;

manhattan += Math.abs(diff);

dot += A[i]*B[i];

normA += A[i]*A[i];

normB += B[i]B[i];
}
```

```
euclidean =
Math.sqrt(euclidean);

normA =
Math.sqrt(normA);

normB =
Math.sqrt(normB);

let cosine =
dot /
(normAnormB);

cosine =
Math.max(
-1,
Math.min(
1,
cosine
)
);

let angle =
Math.acos(cosine);

let angleDeg =
angle *
180 /
Math.PI;

document
.getElementById(
"results"
)
.innerHTML =
```

```
</span><br></div><div><span
attribution="{&quot;name&quot;:&quot;Copilot&quot;,&quot;oid&quot;:&quot;E
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quot;timestamp&quot;:1793397600000,&quot;dataSource&quot;:0}">&lt;h3&gt;Re
sults&lt;/h3&gt;</span><br></div><div><span
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{euclidean.toFixed(6)}
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</div><div>Manhattan Distance:
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$< /span >< br >< /div >< div >< spanattribution = ""name":&quot$
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```

MathJax.typesetPromise();
}

</script>

---

5. Add a "Dimensionality Explorer"

This helps beginners understand how geometry changes with dimensionality.

<div
class="term"

```
onclick="toggle(this)">
Dimensionality Explorer
</div>

<div class="content">

<input
type="range"
min="1"
max="1000"
value="3"
id="dimSlider"
oninput="updateDimension()">

<div id="dimOutput"></div>

</div>
</div>
```

Add:

```
function updateDimension()
{
let d =
document
.getElementById(
"dimSlider"
).value;

let text = "";

if(d==1)
text =
"A line";

else if(d==2)
text =
"A plane";
```

```
else if(d==3)
text =
"Three-dimensional space";

else if(d<10)
text =
"Low-dimensional geometry";

else if(d<100)
text =
"Moderately high-dimensional space";

else if(d<1000)
text =
"Embedding-like dimensionality";

else
text =
"Extremely high-dimensional space";

document
.getElementById(
"dimOutput"
)
.innerHTML =

&lt;h3&gt;
dDimensions&lt;/h3&gt;< /span >< br >< /div >< div >< spanattribution = "&quot;name&quot;:&quot;Co
{text}&lt;/p&gt;;
}

updateDimension();
```

---

6. Add a Formula Glossary Section

<div
class="term"
onclick="toggle(this)">
Formula Reference Sheet

[
\textbf{Euclidean Distance}
]

[
d=
\sqrt{\sum_{i=1}^{n}
(x_i-y_i)^2}
]

<hr>

[
\textbf{Manhattan Distance}
]

[
d=
\sum_{i=1}^{n}
|x_i-y_i|
]

<hr>

[
\textbf{Dot Product}
]

$$
x \cdot y
=
\sum_{i=1}^{n}
x_i y_i
$$

<hr>

$$
\textbf{Cosine Similarity}
$$

$$
\cos(\theta)
=
\frac{x \cdot y}
{|x| |y|}
$$

<hr>

$$
\textbf{Angle}
$$

$$
\theta =
\arccos(
\cos(\theta)
)
$$

Result

Your glossary evolves from a static reference into an interactive learning environment where a user can:

- Explore dimensions from **1D to 1000D+**
- Enter arbitrary vectors
- Calculate **Euclidean distance**
- Calculate **Manhattan distance**
- Calculate **dot products**
- Calculate **cosine similarity**
- Calculate **angles between vectors**
- View properly rendered mathematical notation with **MathJax**
- Build intuition for measurements in high-dimensional spaces.

Good idea. The current calculator code has several edge cases that can produce incorrect results, `NaN`, `Infinity`, or misleading output.

Validation Issues in the Current Version

1. Empty Inputs

Current code:

parseFloat(
document.getElementById(`a${i}`).value
)

Problem:

"" → NaN
" " → NaN

This propagates through all calculations.

---

2. Non-Numeric Inputs

Examples:

abc
1,2
∞
hello

Result:

NaN

which breaks every metric.

---

3. Null Elements

If the DOM changes:

document.getElementById("a5")

may return:

null
``

causing:

Cannot read properties of null

---

4. Zero-Vector Problems

Very important.

Example:

A = [0,0,0]

B = [1,2,3]

Norm:

||A|| = 0

Cosine similarity becomes:

00\frac{0}{0}

which is undefined.

Current code:

dot/(normA*normB)

becomes:

0 / 0

or

5 / 0

causing:

NaN
Infinity

---

5. Dimension Mismatch

Future versions might allow pasted vectors:

A = [1,2,3]

B = [1,2]
`

Those dimensions do not match.

---

6. Extremely Large Values

Example:

1e308

Squaring can overflow:

Infinity

leading to meaningless distances.

---

7. Negative Dimension

User manual edit:

<input value="-50">

Need validation.

---

8. Non-Integer Dimension

Current input allows:

3.14

Dimensions must be integer.

Recommended Validation Layer

Add:

```
function validateDimension()
{
const n = Number(
document.getElementById(
"dimension"
).value
);

if(!Number.isInteger(n))
{
```

```
throw Error(
"Dimension must be an integer."
);
}

if(n < 1)
{
throw Error(
"Dimension must be at least 1."
);
}

if(n > 10000)
{
throw Error(
"Dimension exceeds supported range."
);
}

return n;
}
```

Safe Number Reader

```
function readNumber(id)
{
const element =
document.getElementById(id);

if(!element)
{
throw Error(
Missing input: ${id}
);
}
```

```
const value =
element.value.trim();

if(value === "")
{
throw Error(
`${id} is empty`
);
}

const num =
Number(value);

if(!Number.isFinite(num))
{
throw Error(
`${id} is not a valid number`
);
}

return num;
}
```

Zero-Vector Detection

```
function isZeroVector(v)
{
const EPS = 1e-12;

return v.every(
x => Math.abs(x) < EPS
);
}
```

Usage:

```
if(isZeroVector(A))
{
throw Error(
"Vector A cannot be a zero vector when computing angles or cosine similarity."
);
}

if(isZeroVector(B))
{
throw Error(
"Vector B cannot be a zero vector when computing angles or cosine similarity."
);
}
```

Near-Zero Vector Detection

Better:

```
function vectorNorm(v)
{
return Math.sqrt(
v.reduce(
(s,x)=>s+x*x,
0
)
);
}
```

Then:

```
const EPS = 1e-12;

if(vectorNorm(A)<EPS)
{
...
}
```

This catches:

[0,0,0]

[1e-20,-1e-20,1e-21]

which is numerically a zero vector.

Safe Cosine Similarity

Replace:

```
let cosine =
dot /
(normA*normB);
```

With:

```
const EPS = 1e-12;

if(normA < EPS || normB < EPS)
{
cosine = null;
}
else
{
cosine =
dot /
(normA*normB);

cosine =
Math.max(
-1,
Math.min(
1,
cosine
```

)
);
}

Why clamp?

Floating-point calculations sometimes create:

1.0000000001

which causes:

Math.acos(...)

to return:

NaN

Safe Angle Handling

let angleDeg = null;

if(cosine !== null)
{
angleDeg =
Math.acos(cosine)
*180/Math.PI;
}

Display:

const angleText =
angleDeg===null
? "Undefined (zero-vector)"
: angleDeg.toFixed(6)+"°";
``

Sanity Check Vector Sizes

Before calculations:

```
if(A.length !== B.length)
{
throw Error(
"Dimension mismatch."
);
}
```

Overflow Protection

```
const LIMIT = 1e150;

for(const x of [...A,...B])
{
if(Math.abs(x) > LIMIT)
{
throw Error(
"Value too large."
);
}
}
```

User-Friendly Error Display

Wrap the calculation:

```
try
{
calculateMetricsSafe();
}
```

```
catch(err)
{
document
.getElementById("results")
.innerHTML =
</span><br></div><div><span
attribution="{&quot;name&quot;:&quot;Copilot&quot;,&quot;oid&quot;:&quot;E
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ot;Copilot&quot;,&quot;oid&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00
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quot;timestamp&quot;:1793397600000,&quot;dataSource&quot;:0}"> &lt;div
class="result-box"</span><br></div><div><span
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12-4514-AD9B-893A6FAFD00C&quot;,&quot;userInfo&quot;:{&quot;name&quot;:&qu
ot;Copilot&quot;,&quot;oid&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00
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&lt;b&gt;Error:&lt;/b&gt;</span><br></div><div><span
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12-4514-AD9B-893A6FAFD00C&quot;,&quot;userInfo&quot;:{&quot;name&quot;:&qu
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quot;timestamp&quot;:1793397600000,&quot;dataSource&quot;:0}">
```

```
${err.message}</span><br></div><div><span
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12-4514-AD9B-893A6FAFD00C&quot;,&quot;userInfo&quot;:{&quot;name&quot;:&qu
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quot;timestamp&quot;:1793397600000,&quot;dataSource&quot;:0}">
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12-4514-AD9B-893A6FAFD00C&quot;,&quot;userInfo&quot;:{&quot;name&quot;:&qu
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C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;},&
quot;timestamp&quot;:1793397600000,&quot;dataSource&quot;:0}"> ;
}
```

Additional Educational Warnings

For a learning environment, show warnings rather than just blocking:

Orthogonal Vectors

if(Math.abs(cosine) < 1e-6)

Display:

Vectors are approximately orthogonal (90° apart).

Parallel

if(Math.abs(cosine-1) < 1e-6)

Display:

Vectors are approximately parallel.

Antiparallel

if(Math.abs(cosine+1) < 1e-6)

Display:

Vectors are approximately opposite.

Recommended Additional Metrics

Since this is becoming a geometry-learning tool, consider adding validation and computation for:

✓ Euclidean Distance
✓ Manhattan Distance
✓ Dot Product
✓ Cosine Similarity
✓ Angle
✓ Vector Norms
✓ Unit Vectors
✓ Projection of A onto B
✓ Cross Product (3D only)
✓ Mahalanobis Distance
✓ Minkowski Distance (user-selectable p)
✓ Hyperbolic Distance
✓ Spherical Distance

These additions would make the glossary significantly more robust and mathematically correct, particularly around the important edge case of null inputs and zero vectors where cosine similarity and angle calculations are fundamentally undefined.

Yes. Replace the entire `calculateMetrics()` implementation with the hardened version below and add the helper functions. This version handles:

- ✅ Empty inputs
- ✅ Null DOM elements
- ✅ Non-numeric values
- ✅ Infinite values
- ✅ Invalid dimensions

- ✅ Dimension mismatch
- ✅ Zero vectors
- ✅ Near-zero vectors
- ✅ Floating-point acos errors
- ✅ Very large values
- ✅ Educational warnings (parallel, orthogonal, opposite)

Add Helper Functions

function showError(message)
{
document.getElementById("results").innerHTML =

 <span
attribution="{"name":"Copilot","oid":"E
64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E
12-4514-AD9B-893A6FAFD00C","userInfo":{"name":&qu
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C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},&
quot;timestamp":1793397600000,"dataSource":0}"> <div
class="result-box"
 <span
attribution="{"name":"Copilot","oid":"E
64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E
12-4514-AD9B-893A6FAFD00C","userInfo":{"name":&qu
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quot;timestamp":1793397600000,"dataSource":0}">
style="
 <span
attribution="{"name":"Copilot","oid":"E
64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E
12-4514-AD9B-893A6FAFD00C","userInfo":{"name":&qu
ot;Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00
C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},&
quot;timestamp":1793397600000,"dataSource":0}">
background:#ffe6e6;
 <span

attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397600000,"dataSource":0}"> border:1px solid #cc0000;
</div><div> ">
</div><div> <h3>Error</h3>
</div><div> <p>${message}</p>
</div><div><span attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},&

```
quot;timestamp&quot;:1793397600000,&quot;dataSource&quot;:0}">
&lt;/div&gt;</span><br></div><div><span
attribution="{&quot;name&quot;:&quot;Copilot&quot;,&quot;oid&quot;:&quot;E
64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E
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ot;Copilot&quot;,&quot;oid&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00
C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;},&
quot;timestamp&quot;:1793397600000,&quot;dataSource&quot;:0}"> ;
}

function validateDimension()
{
const n = Number(
document.getElementById("dimension").value
);

if(!Number.isInteger(n))
{
throw Error(
"Dimension must be an integer."
);
}

if(n < 1)
{
throw Error(
"Dimension must be at least 1."
);
}

if(n > 10000)
{
throw Error(
"Dimension exceeds supported limit."
);
}
```

```
return n;
}

function readNumber(id)
{
const element =
document.getElementById(id);

if(!element)
{
throw Error(
Missing input: ${id}
);
}

const raw =
element.value.trim();

if(raw === "")
{
throw Error(
${id} cannot be empty.
);
}

const num = Number(raw);

if(!Number.isFinite(num))
{
throw Error(
${id} is not a valid finite number.
);
}

return num;
}
```

```
function vectorNorm(v)
{
return Math.sqrt(
v.reduce(
(sum,x)=>sum+x*x,
0
)
);
}

function isNearlyZeroVector(v)
{
return (
vectorNorm(v) < 1e-12
);
}
```

Replace `calculateMetrics()`

```
function calculateMetrics()
{
try
{
const n =
validateDimension();

const A = [];
const B = [];

for(let i=0;i<n;i++)
{
const a =
readNumber(a${i});

const b =
readNumber(b${i});
```

```
if(Math.abs(a) > 1e150)
{
throw Error(
a${i} is too large.
);
}

if(Math.abs(b) > 1e150)
{
throw Error(
b${i} is too large.
);
}

A.push(a);
B.push(b);
}

if(A.length !== B.length)
{
throw Error(
"Dimension mismatch."
);
}

let euclideanSquared = 0;
let manhattan = 0;
let dot = 0;

for(let i=0;i<n;i++)
{
const diff =
A[i] - B[i];

euclideanSquared +=
diff * diff;
```

```
manhattan +=
Math.abs(diff);

dot +=
A[i] * B[i];
}

const normA =
vectorNorm(A);

const normB =
vectorNorm(B);

const euclidean =
Math.sqrt(
euclideanSquared
);

let cosine = null;
let angleDeg = null;

const aZero =
isNearlyZeroVector(A);

const bZero =
isNearlyZeroVector(B);

if(!(aZero || bZero))
{
cosine =
dot /
(normA * normB);

cosine =
Math.max(
-1,
```

```
Math.min(
1,
cosine
)
);

angleDeg =
Math.acos(cosine)
*
180
/
Math.PI;
}

let warning = "";

if(cosine !== null)
{
if(Math.abs(cosine) < 1e-6)
{
warning =
</span><br></div><div><span
attribution="{&quot;name&quot;:&quot;Copilot&quot;,&quot;oid&quot;:&quot;E
64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E
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quot;timestamp&quot;:1793397600000,&quot;dataSource&quot;:0}">
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quot;timestamp&quot;:1793397600000,&quot;dataSource&quot;:0}"> ✅ Vectors
```

```
are approximately</span><br></div><div><span
attribution="{&quot;name&quot;:&quot;Copilot&quot;,&quot;oid&quot;:&quot;E
64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E
12-4514-AD9B-893A6FAFD00C&quot;,&quot;userInfo&quot;:{&quot;name&quot;:&qu
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C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;},&
quot;timestamp&quot;:1793397600000,&quot;dataSource&quot;:0}"> orthogonal
(90° apart).</span><br></div><div><span
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quot;timestamp&quot;:1793397600000,&quot;dataSource&quot;:0}">
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C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;},&
quot;timestamp&quot;:1793397600000,&quot;dataSource&quot;:0}"> ;
}
else if(
Math.abs(cosine - 1)
< 1e-6
)
{
warning =
</span><br></div><div><span
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64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E
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```

```
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12-4514-AD9B-893A6FAFD00C&quot;,&quot;userInfo&quot;:{&quot;name&quot;:&qu
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C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;},&
quot;timestamp&quot;:1793397600000,&quot;dataSource&quot;:0}"> ✅ Vectors
are approximately</span><br></div><div><span
attribution="{&quot;name&quot;:&quot;Copilot&quot;,&quot;oid&quot;:&quot;E
64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E
12-4514-AD9B-893A6FAFD00C&quot;,&quot;userInfo&quot;:{&quot;name&quot;:&qu
ot;Copilot&quot;,&quot;oid&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00
C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;},&
quot;timestamp&quot;:1793397600000,&quot;dataSource&quot;:0}">
parallel.</span><br></div><div><span
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12-4514-AD9B-893A6FAFD00C&quot;,&quot;userInfo&quot;:{&quot;name&quot;:&qu
ot;Copilot&quot;,&quot;oid&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00
C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;},&
quot;timestamp&quot;:1793397600000,&quot;dataSource&quot;:0}">
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12-4514-AD9B-893A6FAFD00C&quot;,&quot;userInfo&quot;:{&quot;name&quot;:&qu
ot;Copilot&quot;,&quot;oid&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00
C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;},&
quot;timestamp&quot;:1793397600000,&quot;dataSource&quot;:0}"> ;
}
else if(
Math.abs(cosine + 1)
```

< 1e-6
)
{
warning =

 <span
attribution="{"name":"Copilot","oid":"E
64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E
12-4514-AD9B-893A6FAFD00C","userInfo":{"name":&qu
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quot;timestamp":1793397600000,"dataSource":0}">
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C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},&
quot;timestamp":1793397600000,"dataSource":0}"> ✅ Vectors
are approximately
 <span
attribution="{"name":"Copilot","oid":"E
64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E
12-4514-AD9B-893A6FAFD00C","userInfo":{"name":&qu
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quot;timestamp":1793397600000,"dataSource":0}">
opposite.
 <span
attribution="{"name":"Copilot","oid":"E
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quot;timestamp":1793397600000,"dataSource":0}">
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 <span

```
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64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E
12-4514-AD9B-893A6FAFD00C&quot;,&quot;userInfo&quot;:{&quot;name&quot;:&qu
ot;Copilot&quot;,&quot;oid&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00
C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;},&
quot;timestamp&quot;:1793397600000,&quot;dataSource&quot;:0}"> ;
}
}

const cosineDisplay =
cosine === null
? "Undefined (zero vector)"
: cosine.toFixed(8);

const angleDisplay =
angleDeg === null
? "Undefined (zero vector)"
: angleDeg.toFixed(6) + "°";

document.getElementById(
"results"
).innerHTML =
</span><br></div><div><span
attribution="{&quot;name&quot;:&quot;Copilot&quot;,&quot;oid&quot;:&quot;E
64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E
12-4514-AD9B-893A6FAFD00C&quot;,&quot;userInfo&quot;:{&quot;name&quot;:&qu
ot;Copilot&quot;,&quot;oid&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00
C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;},&
quot;timestamp&quot;:1793397600000,&quot;dataSource&quot;:0}"> &lt;div
class="result-box"&gt;</span><br></div><div><span
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```
if(window.MathJax)
{
MathJax.typesetPromise();
}
}
catch(error)
{
showError(error.message);
}
}
```

Optional Improvement for High Dimensions

For dimensions above ~100, rendering thousands of input boxes becomes unusable. Replace manual coordinate entry with two text areas:

1,2,3,4,5,6

and

7,8,9,10,11,12

then parse them into vectors. This scales cleanly to:

- 128 dimensions
- 384 dimensions
- 768 dimensions
- 1536 dimensions
- 4096 dimensions

which is much more practical for embedding-space exploration.

For large dimensions (100-100,000+), the biggest bottleneck is not the math. It's the DOM and repeated memory allocations.

Key optimizations

1. Switch from thousands of input boxes to textarea vectors

Instead of:

<input id="a0">
<input id="a1">
...
<input id="a9999">

use:

<textarea id="vectorA"></textarea>
<textarea id="vectorB"></textarea>

Example input:

1,2,3,4,5

or

1
2
3
4
5

This scales much better.

---

2. Single-pass metric computation

Instead of multiple loops:

for(...) euclidean...
for(...) dot...
for(...) norms...

compute everything in one pass:

```
function computeMetrics(A, B)
{
const n = A.length;

let dot = 0;
let normA2 = 0;
let normB2 = 0;

let manhattan = 0;
let euclidean2 = 0;

for(let i=0;i<n;i++)
{
const a = A[i];
const b = B[i];

const d = a - b;

dot += a * b;

normA2 += a * a;
normB2 += b * b;

manhattan += Math.abs(d);
euclidean2 += d * d;
}
```

```
return {
dot,
normA: Math.sqrt(normA2),
normB: Math.sqrt(normB2),
manhattan,
euclidean: Math.sqrt(euclidean2)
};
}
```

Complexity:

O(n)

with only one traversal.

---

3. Use Typed Arrays

For high-dimensional embeddings:

```
const A = new Float64Array(valuesA);
const B = new Float64Array(valuesB);
```

Benefits:

- Lower memory overhead
- Faster numeric loops
- Better cache locality

Example parser:

```
function parseVector(text)
{
const values =
text
.trim()
.split(/[\s,]+/)
.map(Number);

return new Float64Array(values);
}
```

---

4. Fast norm detection

Instead of:

vectorNorm(v)
`

then another pass

use accumulated norms:

const EPS = 1e-12;

const normA =
Math.sqrt(normA2);

const normB =
Math.sqrt(normB2);

const zeroA =
normA < EPS;

const zeroB =
normB < EPS;

No extra work required.

---

5. Chunked processing for huge vectors

For:

100k+
1m+

dimensions, allow the UI to update:

async function computeLarge(A, B)

```
{
const chunk = 10000;

let accum = {...};

for(let start=0;
start<A.length;
start+=chunk)
{
const end =
Math.min(
start+chunk,
A.length
);

for(let i=start;i<end;i++)
{
...
}

await new Promise(
r => setTimeout(r,0)
);
}

return accum;
}
```

Prevents browser freezes.

---

6. Add Minkowski distances efficiently

Instead of separate implementations:

```
function minkowski(A,B,p)
{
let sum = 0;
```

```
for(let i=0;i<A.length;i++)
{
sum +=
Math.pow(
Math.abs(A[i]-B[i]),
p
);
}

return Math.pow(sum,1/p);
}
```

Provides:

p=1 Manhattan
p=2 Euclidean
p=∞ Chebyshev

from a single implementation.

---

7. Web Worker for very large dimensions

For 10k-1M dimensions:

```
worker.postMessage({
A,
B
});
```

Move calculations into:

worker.js

Benefits:

- UI remains responsive
- Progress bars become possible
- Future UMAP playground won't freeze

---

8. Add numerical stability

For embeddings, norms can become large.

Use compensated summation:

```
function kahanSum(values)
{
let sum = 0;
let c = 0;

for(const x of values)
{
const y = x - c;
const t = sum + y;

c = (t - sum) - y;
sum = t;
}

return sum;
}
```

Useful for:

768D
1536D
4096D
100000D

vectors.

Recommended architecture

Use three modes:

Small (<100D)

Individual input boxes

Good for learning.

Medium (100-10,000D)

Textarea vectors

Good for experimentation.

Large (>10,000D)

Textarea
+
Typed Arrays
+
Web Worker

Good for embedding research.

Add practical embedding support

A particularly useful addition for your glossary would be:

<button>
Generate Random 768D Embedding
</button>

<button>
Generate Random 1536D Embedding
</button>

<button>
Generate Random 4096D Embedding
</button>

Then compute:

- Euclidean distance
- Manhattan distance
- Cosine similarity
- Angle
- Norms

using the optimized single-pass implementation.

That closely matches real-world vector sizes used by modern embedding models and lets learners immediately explore how distance and angle behave as dimensionality grows.

To make the glossary a **self-contained learning kit from absolute beginner to advanced practitioner**, I'd expand it from a collection of definitions into a structured knowledge system.

1. Add a "Mathematical Language" Section

Many learners struggle because they don't know what symbols mean.

Symbols Glossary

Equals

$

Means:>"hasthesamevalueas"Example: Means: > "has the same value as" Example:

2+3=5

*** ### Plus

+

Addition.Example: Addition. Example:

3+4=7

*** ### Minus ## $

Subtraction.

Example:

7−2=57-2=5

Multiply

×\times

or

⋅\cdot

Example:

3×4=123 \times 4 = 12

Divide

ab\frac{a}{b}

Means:

a divided by b

Square

x2x^2

Means:

x×xx\times x

Example:

32=93^2 = 9

Square Root

x\sqrt{x}

Means:

> what number multiplied by itself gives x?

Example:

9=3\sqrt{9}=3

Summation

∑\sum

Means:

> add everything together

Example:

\sum_{i=1}^{3} i ================= # 1+2+3 6

Absolute Value

|x||x|

Distance from zero.

Examples:

// −5|=5\|-5|=5 |5|=5|5|=5

Infinity

∞\infty

Not a number.

Represents unbounded growth.

---

2. Build Formula Decomposition Panels

Every formula should be clickable.

Instead of:

d=∑_i=1n(x_i−y_i)2d= \sqrt{ \sum_{i=1}^{n} (x_i-y_i)^2 }

show:

Euclidean Distance

Meaning:

> Straight-line distance

Formula:

d=∑_i=1n(x_i−y_i)2d= \sqrt{ \sum_{i=1}^{n} (x_i-y_i)^2 }

Where:

Symbol	Meaning
dd	distance
x_ix_i	coordinate of first point
y_iy_i	coordinate of second point
nn	number of dimensions
∑\sum	add all dimensions
\sqrt{}	square root

Add a "Show Worked Example" button:

A=(1,2)

B=(4,6)

Display:

(4-1)^2 = 9

(6-2)^2 = 16

9+16 = 25

√25 = 5

---

3. Add Metric Encyclopedia

Current glossary barely explains metrics.

Metrics deserve their own category.

Euclidean Distance

Intuition

Crow flies directly.

A ------ B

Formula:

∑(x_i−y_i)2\sqrt{ \sum (x_i-y_i)^2 }

Best for:

- physics
- geometry
- PCA

Manhattan Distance

Intuition

Walking city blocks.

+---+

| |

+---+

Formula:

∑∣x_i−y_i∣\sum |x_i-y_i|

Best for:

- grids
- sparse data

Chebyshev Distance

Intuition

Maximum coordinate difference.

Formula:

max_i∣x_i−y_i∣\max_i |x_i-y_i|

Useful for:

- chess king movement

Minkowski Distance

A family of distances.

Formula:

(∑∣x_i−y_i∣p)1/p\left(\sum |x_i-y_i|^p \right)^{1/p}

Special cases:

p=1 Manhattan

p=2 Euclidean

p→∞ Chebyshev

Cosine Similarity

Measures Direction

Not distance.

Formula:

cos(θ)============x·y∣∣x∣∣∣∣y∣∣\cos(\theta) ============ \frac{ x\cdot y }{ ||x|| ||y|| }

Values:

1 same direction

0 orthogonal

-1 opposite

Useful for:

- embeddings
- search
- NLP

Cosine Distance

Formula:

1−cos(θ)1-\cos(\theta)

Shows directional difference.

Hamming Distance

Counts differences

101100

101000

Difference count:

1

Useful for:

- binary strings
- error correction

Jaccard Similarity

Measures overlap.

Formula:

∣A∩B∣∣A∪B∣\frac{ |A\cap B| }{ |A\cup B| }

Example:

A={cat,dog}

B={dog,wolf}

Overlap:

1/3

Mahalanobis Distance

Distance adjusted for covariance.

Useful when dimensions have:

different scales

correlations

Can detect outliers better than Euclidean distance.

Haversine Distance

Distance along a sphere.

Used for:

Earth
GPS
Latitude/Longitude

Hyperbolic Distance

Distance in negatively curved space.

Useful for:

hierarchies
trees
taxonomies
graphs

---

4. Add Geometry Encyclopedia

Euclidean Geometry

Flat space.

Examples:

paper
floor
screen

Curvature:

00

Spherical Geometry

Surface of a sphere.

Examples:

Earth
Moon
Planets

Curvature:

Positive.

Hyperbolic Geometry

Space expands exponentially.

Curvature:

Negative.

Useful for representing:

trees
language
knowledge graphs

---

5. Add Linear Algebra Encyclopedia

Current glossary needs deeper foundations.

Matrix

Visual:

[
1 2
3 4
]

A rectangular grid of numbers.

Transpose

ATA^T

Rows become columns.

Identity Matrix

II

Matrix equivalent of 1.

Eigenvector

Special direction that remains aligned after transformation.

Eigenvalue

Scaling factor applied to an eigenvector.

Covariance

Measures how variables move together.

Positive:

both increase

Negative:

one increases
one decreases

---

6. Add High-Dimensional Intuition Panels

Most learners struggle here.

Why 768 Dimensions Isn't Magic

A vector:

[1,4,2,8,...]
``

is no different mathematically from:

[1,4]

The formulas stay the same.

Only the number of coordinates changes.

Curse of Dimensionality

As dimensions increase:

- distances become less informative
- neighborhoods become sparse
- intuition breaks down

Add an interactive simulation.

Concentration of Distances

Show that:

10D

100D

1000D

produce increasingly similar pairwise distances.

Huge insight for ML practitioners.

---

7. Add Interactive Derivations

For every major formula:

Button:

Show Derivation

Examples:

- Pythagorean theorem
- Euclidean distance
- Cosine similarity
- Hyperboloid equation
- Poincaré projection
- UMAP objective

---

8. Add Knowledge Prerequisite Graph

Each term gets:

Requires:

Example:

Cosine Similarity

Requires:

- Vector
- Dot Product
- Norm
- Angle

Clickable links.

---

9. Add Learning Paths

Path A

Beginner

Point
Coordinate
Dimension
Distance
Angle
Vector

Path B

Machine Learning

Vector
Norm
Dot Product
Cosine Similarity
Embedding
UMAP

Path C

Hyperbolic Geometry

Curvature
Manifold
Hyperbolic Space
Hyperboloid
Poincaré Disk
Geodesic

---

10. Add a "Can You Explain This Like I'm 10?" Toggle

Every entry gets:

Technical

Formal mathematical definition

Intuitive

Simple explanation

Example:

Hyperbolic Space (technical)

> A manifold with constant negative curvature.

Hyperbolic Space (age 10 version)

> Imagine a world where every step outward creates more room than you expect, so tree-like structures fit naturally.

With these additions, the glossary stops being a reference page and becomes a complete interactive curriculum covering:

- arithmetic
- algebra
- geometry
- trigonometry
- linear algebra
- metric spaces
- manifolds
- dimensionality reduction
- embeddings
- UMAP
- spherical geometry
- hyperbolic geometry

- high-dimensional reasoning

all within a single self-contained learning environment.

Interactive formula walkthroughs are one of the highest-value additions you can make. Instead of showing a formula, you guide the learner through it one symbol at a time, with explanations, visualizations, and worked examples.

Design Goals

Each walkthrough should provide:

1. **What problem does this formula solve?**
2. **What does every symbol mean?**
3. **Worked example**
4. **Step-by-step reveal**
5. **Animation**
6. **Interactive input**
7. **Common mistakes**
8. **Visual interpretation**
9. **Dimensionality scaling**

Walkthrough Framework

Each formula card becomes:

[Formula Name]

What does it do?

[Step 1]
[Step 2]
[Step 3]
...

[Interactive Example]

[Check My Understanding]

Example 1: Euclidean Distance Walkthrough

Initial Screen

d=
\sqrt{
\sum_{i=1}^{n}
(x_i-y_i)^2
}
``

Question:

> How far apart are two points?

Step 1

Highlight:

x_i-y_i

Explanation:

> Find the difference between the coordinates.

Example

A = (1,2)

B = (4,6)

Dimension-by-dimension:

4 - 1 = 3

6 - 2 = 4

Visual:

(1,2) •

• (4,6)

Step 2

Highlight:

(x_i-y_i)^2

Explanation:

Square every difference so negative values don't cancel out positives.

Example:

$3^2 = 9$

$4^2 = 16$

Step 3

Highlight:

Σ

Explanation:

Add all squared differences.

Example:

9 + 16 = 25

Step 4

Highlight:

√

Explanation:

Take the square root to recover the distance scale.

Example:

√25 = 5

Final

Answer:

Distance = 5

Visual:

A •----------• B

Length 5

HTML Structure

[
d=\sqrt{

\sum(x_i-y_i)^2
}
]

Calculate differences.

<button onclick="nextStep()">
Next Step
</button>

Walkthrough 2: Dot Product

Formula:

x \cdot y
=
\sum x_i y_i

Step 1

Multiply matching coordinates:

[1,2]

[3,4]

Produces:

1×3

2×4
``

Step 2

Add results:

3 + 8 = 11

Visual Meaning

Positive:

→ →
``

Same direction.

Zero:

→
↑

Perpendicular.

Negative:

→ ←

Opposite direction.

Walkthrough 3: Cosine Similarity

Formula:

\cos(\theta)
=
\frac
{x\cdot y}
{||x||,||y||}
``

Step 1

Compute dot product.

Numerator

Step 2

Compute vector lengths.

Denominator

Step 3

Divide.

cos θ

Step 4

Interpret result.

+1

Same direction
`

0

Perpendicular

-1

Opposite direction

Interactive Angle Wheel

Create a visual gauge:

-1 ----- 0 ----- +1

← ↑ →

Move a slider:

<input
type="range"
min="-1"
max="1"
step="0.01">

Display:

Cosine = 0.73

Angle = 43°

Walkthrough 4: Manhattan Distance

Formula:

\sum |x_i-y_i|

Visual:

City blocks

+---+
| |
+---+

Interactive:

Show two points.

Compute:

Street distance

versus

Direct distance

Walkthrough 5: Minkowski Distance

Formula:

\left(
\sum
|x_i-y_i|^p
\right)^{1/p}

Add slider:

p = 1

Then animate:

Manhattan

Move to:

p = 2

Show:

Euclidean

Move toward:

p = 100

Show:

Approaching Chebyshev

This creates tremendous geometric intuition.

Walkthrough 6: Hyperboloid Formula

Formula:

z^2 - x^2 - y^2 = 1

Step 1

Explain:

Ordinary sphere:

$x^2+y^2+z^2=1$
`

Step 2

Explain sign change:

Hyperboloid:

$z^2-x^2-y^2=1$

Step 3

Render live 3D model.

Rotate.

Highlight selected point.

Step 4

Ask:

What happens if x grows?

Animate.

Walkthrough 7: Poincaré Projection

Formula:

$$X=\frac{x}{1+z}$$
$$Y=\frac{y}{1+z}$$

Interactive

Show:

Hyperboloid

on left.

Poincaré disk on right.

Slider:

Move point

Watch projection update live.

Walkthrough 8: UMAP Metric Walkthrough

Goal:

Teach input metric vs output metric visually.

Panel 1

Show points.

Panel 2

Select:

Euclidean
Cosine
Jaccard
``

Input metric.

Panel 3

Watch neighbor graph update.

Panel 4

Select:

Euclidean
Sphere
Hyperboloid

Output metric.

Watch embedding geometry change.

Knowledge Check Component

After each walkthrough:

Question:

For Euclidean distance:

A=(1,2)

B=(4,6)

What is the distance?

Learner enters:

5

Immediate feedback.

Progressive Disclosure

Add:

Show Simple Explanation

and

Show Technical Explanation

Example:

Simple

> Distance tells us how far apart two things are.

Technical

> Euclidean distance is the L_2 norm of the difference vector.

Final Result

Your glossary becomes:

- Interactive encyclopedia
- Formula tutor
- Geometry calculator
- Metric explorer
- Hyperbolic-space visualizer
- UMAP learning environment
- Self-paced course from arithmetic through high-dimensional embeddings

rather than a static glossary.

A reusable component should be **data-driven**, not hardcoded to a specific formula. The formula, steps, examples, explanations, quiz questions, and visualizations should all be supplied as configuration.

Design Goals

One component should support:

- Euclidean Distance
- Manhattan Distance
- Dot Product
- Cosine Similarity
- Minkowski Distance
- Haversine Distance
- Hyperboloid Equation
- Poincaré Projection
- UMAP Input vs Output Metrics
- Future formulas

without changing the rendering code.

Formula Definition Schema

const euclideanDistance = {
id: "euclidean-distance",

title: "Euclidean Distance",

category: "Metrics",

purpose:
"Measures straight-line distance between two points.",

formula:
 <span attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397600000,"dataSource":0}"> d =
 <span attribution="{"name":"Copilot","oid":"E

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\\sqrt{
 <span
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\\sum_{i=1}^{n}
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(x_i-y_i)^2
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symbols: [
{
symbol: "d",
meaning: "distance"
},
{
symbol: "x_i",
meaning: "coordinate of first point"
},
{
symbol: "y_i",
meaning: "coordinate of second point"
},
{
symbol: "\sum",
meaning: "sum over all dimensions"
},
{
symbol: "\sqrt{}",
meaning: "square root"
}
],

steps: [
{
title: "Find Differences",

explanation:
"Subtract coordinates dimension-by-dimension.",

formula:
"x_i-y_i"
},

{
title: "Square Differences",

explanation:
"Square each difference to eliminate sign.",

formula:
"(x_i-y_i)^2"
},

{
title: "Add Results",

explanation:
"Sum all squared differences.",

formula:
"\sum (x_i-y_i)^2"
},

{
title: "Square Root",

explanation:
"Return to distance units.",

formula:
"\sqrt{\cdots}"
}
],

example: {
input: {
A: [1,2],
B: [4,6]
},

walkthrough: [
"Differences: [3,4]",

"Squares: [9,16]",
"Sum: 25",
"Distance: 5"
]
},

quiz: {
question:
"What is the distance between (0,0) and (3,4)?",

answer: 5
}
};

Generic Component

Renderer

class FormulaWalkthrough
{
constructor(containerId)
{
this.container =
document.getElementById(
containerId
);
}

render(definition)
{
this.definition = definition;
this.currentStep = 0;

this.container.innerHTML =

```
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ot;Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00
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 <span attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397600000,"dataSource":0}"> ${step.formula}
 <span attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00

C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397600000,"dataSource":0}"> \\]
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 <span attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397600000,"dataSource":0}"> <button
 <span attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397600000,"dataSource":0}"> onclick="
 <span attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E

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 <span attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397600000,"dataSource":0}">Previous
 <span attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397600000,"dataSource":0}"></button>
 <span attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397600000,"dataSource":0}">
 <span attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397600000,"dataSource":0}"><button
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onclick="
 <span
attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397600000,"dataSource":0}">
walkthrough.nextStep()">
 <span
attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397600000,"dataSource":0}">
Next
 <span
attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397600000,"dataSource":0}">
</button>
 <span
attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00

C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397600000,"dataSource":0}"> ;

MathJax.typesetPromise();
}

Navigation

nextStep()
{
if(
this.currentStep <
this.definition.steps.length-1
)
{
this.currentStep++;
}

this.showStep(
this.currentStep
);
}

previousStep()
{
if(this.currentStep>0)
{
this.currentStep--;
}

this.showStep(
this.currentStep
);
}

Example View

```
showExample()
{
const ex =
this.definition.example;

const html =
ex.walkthrough
.map(
x => &lt;li&gt;${x}&lt;/li&gt;
)
.join("");

document
.getElementById(
"walkthroughBody"
).innerHTML =
</span><br></div><div><span
attribution="{&quot;name&quot;:&quot;Copilot&quot;,&quot;oid&quot;:&quot;E
64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E
12-4514-AD9B-893A6FAFD00C&quot;,&quot;userInfo&quot;:{&quot;name&quot;:&qu
ot;Copilot&quot;,&quot;oid&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00
C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;},&
quot;timestamp&quot;:1793397600000,&quot;dataSource&quot;:0}">
&lt;h3&gt;</span><br></div><div><span
attribution="{&quot;name&quot;:&quot;Copilot&quot;,&quot;oid&quot;:&quot;E
64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E
12-4514-AD9B-893A6FAFD00C&quot;,&quot;userInfo&quot;:{&quot;name&quot;:&qu
ot;Copilot&quot;,&quot;oid&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00
C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;},&
quot;timestamp&quot;:1793397600000,&quot;dataSource&quot;:0}">
Example</span><br></div><div><span
attribution="{&quot;name&quot;:&quot;Copilot&quot;,&quot;oid&quot;:&quot;E
```

64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397600000,"dataSource":0}">
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 <span
attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397600000,"dataSource":0}">
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attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397600000,"dataSource":0}">

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${html}
 <span
attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397600000,"dataSource":0}">

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}

Quiz View

showQuiz()
{
document
.getElementById(
"walkthroughBody"
).innerHTML =

 <span attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397600000,"dataSource":0}">
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 <span

attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397600000,"dataSource":0}">
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${this.definition.quiz.question}
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<input
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id="quizAnswer">

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quot;timestamp":1793397600000,"dataSource":0}">
walkthrough.checkQuiz()">
 <span
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quot;timestamp":1793397600000,"dataSource":0}">
Submit
 <span
attribution="{"name":"Copilot","oid":"E
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12-4514-AD9B-893A6FAFD00C","userInfo":{"name":&qu
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</div><div> <div id="quizResult">
</div><div> </div>
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}

Quiz Engine

checkQuiz()
{

```
const value =
Number(
document
.getElementById(
"quizAnswer"
).value
);

const correct =
this.definition.quiz.answer;

document
.getElementById(
"quizResult"
)
.innerHTML =
(
value === correct
)
?
"✅ Correct"
:
"❌ Try Again";
}
```

Enhanced Features

Symbol Highlighting

Clicking a symbol:

Σ

opens:

Meaning:

Adds values together.

and shows an animated example.

Worked Example Animation

For Euclidean distance:

[1,2]
[4,6]

Animate:

3
4

9
16

25

5

step-by-step.

Difficulty Toggle

🟢 Beginner
🔵 Intermediate
🟣 Advanced

Example:

Beginner

Square the difference.

Advanced

Compute the L_2 norm of the difference vector.

Dependency Graph

Each definition can include:

prerequisites: [
"Point",
"Coordinate",
"Vector"
]

Rendered as:

Point
↓
Coordinate
↓
Vector
↓
Euclidean Distance

Interactive Calculator Hook

Each formula may optionally supply:

calculator:
{
enabled:true,

dimensions:true,

vectors:true
}

The component then automatically attaches your validated distance/angle calculator and populates it with the walkthrough example values.

Final Architecture

Formula Registry

↓

FormulaWalkthrough Component

↓

MathJax Renderer

↓

Interactive Example

↓

Quiz Engine

↓

Calculator Engine

↓

Knowledge Graph

This gives you a single reusable component capable of powering the entire glossary while remaining expandable to metrics, linear algebra, manifolds, hyperbolic geometry, embeddings, and UMAP concepts.

To make the glossary feel like a modern learning platform rather than a collection of cards, I recommend a layout with:

- Responsive mobile/tablet/desktop support
- Light + dark mode support
- Sticky navigation
- Formula cards with visual hierarchy
- Interactive walkthrough panels
- MathJax-friendly styling
- Accessible typography
- Knowledge graph styling
- Quiz and calculator panels
- Code/ASCII visualization blocks

Root CSS Variables

```
:root {

--bg: #f5f7fb;
--surface: #ffffff;
--surface-2: #eef4ff;

--text: #1b1b1b;
--text-muted: #666;

--primary: #3366ff;
--primary-hover: #2148c0;

--success: #0a8f47;
--warning: #c98700;
--danger: #c62828;

--border: #d8dce8;

--shadow:
0 4px 12px rgba(0,0,0,.08);

--radius: 14px;

--font:
"Segoe UI",
system-ui,
sans-serif;
}
```

Dark Mode

```
[data-theme="dark"] {
```

--bg: #111827;
--surface: #1f2937;
--surface-2: #253246;

--text: #f5f5f5;
--text-muted: #c4c4c4;

--border: #374151;

--shadow:
0 4px 14px rgba(0,0,0,.45);
}

Dark-mode toggle:

<button onclick="toggleTheme()">
🌙 / ☀️
</button>
function toggleTheme() {
const body = document.body;

body.dataset.theme =
body.dataset.theme === "dark"
? "light"
: "dark";
}

Page Layout

body {

margin: 0;
padding: 0;

background: var(--bg);
color: var(--text);

```
font-family: var(--font);

line-height: 1.6;
}

.app {

display: grid;

grid-template-columns:
280px
1fr;

min-height: 100vh;
}
```

Sidebar Navigation

```
.sidebar {

background: var(--surface);

border-right:
1px solid var(--border);

padding: 20px;

position: sticky;
top: 0;

height: 100vh;

overflow-y: auto;
}
```

```
.sidebar h2 {
margin-top: 0;
}
.sidebar a {

display: block;

padding: 10px;

border-radius: 8px;

text-decoration: none;

color: var(--text);
}

.sidebar a:hover {

background:
var(--surface-2);
}
```

Main Content Area

```
.main {

padding: 30px;

max-width: 1200px;
}
```

Search Box

```
.search-box {
```

width: 100%;

padding: 14px;

font-size: 16px;

border:
1px solid var(--border);

border-radius: 12px;

background:
var(--surface);
}

Generic Card

.card {

background:
var(--surface);

border:
1px solid var(--border);

border-radius:
var(--radius);

box-shadow:
var(--shadow);

padding: 22px;

margin-bottom: 20px;

transition:
.25s ease;
}
``
.card:hover {

transform:
translateY(-2px);
}

Formula Card

.formula-card {

border-left:
6px solid
var(--primary);
}

Category Badge

.category {

display:inline-block;

padding:
4px 10px;

margin-bottom:
10px;

border-radius:
100px;

background:
var(--surface-2);

color:
var(--primary);

font-size:
12px;

font-weight:
600;

text-transform:
uppercase;
}

Formula Display

.formula {

font-size: 1.25rem;

overflow-x: auto;

text-align:center;

padding: 16px;

background:
var(--surface-2);

border-radius: 10px;

margin:

15px 0;
}

MathJax automatically inherits this styling.

Visual Diagrams

.visual {

background:
#0d1117;

color:
#00ff9d;

font-family:
Consolas,
monospace;

padding:
15px;

border-radius:
10px;

overflow:auto;

white-space:pre;
}

Symbol Tables

.symbol-table {

```
width:100%;

border-collapse:
collapse;
}
.symbol-table th,
.symbol-table td {

padding:12px;

border-bottom:
1px solid var(--border);

text-align:left;
}
```

Walkthrough Steps

```
.walkthrough-step {

border-left:
4px solid
var(--primary);

padding-left:
15px;

margin:
20px 0;
}
```

Current step:

```
.walkthrough-step.active {
```

```
background:
var(--surface-2);

border-radius:
8px;

padding:
15px;
}
```

Navigation Buttons

```
.button {

padding:
10px 18px;

border:none;

border-radius:
10px;

cursor:pointer;

background:
var(--primary);

color:white;

transition:
.2s ease;
}
.button:hover {

background:
```

var(--primary-hover);
}

Quiz Styling

.quiz {

background:
var(--surface-2);

border-radius:
12px;

padding:
20px;
}

Correct answer:

.quiz-success {

color:
var(--success);

font-weight:
bold;
}

Wrong answer:

.quiz-error {

color:
var(--danger);

font-weight:
bold;

}

Calculator Panel

.calculator {

background:
var(--surface);

border:
1px solid var(--border);

border-radius:
14px;

padding:
20px;
}

Inputs:

.calculator input,
.calculator textarea,
.calculator select {

width:100%;

padding:12px;

margin-bottom:12px;

border-radius:10px;

border:
1px solid var(--border);

font-size:15px;
}

Result Box

.result-box {

background:
var(--surface-2);

border-left:
5px solid
var(--primary);

padding:
16px;

border-radius:
10px;
}

Metric Comparison Panel

.metric-grid {

display:grid;

grid-template-columns:
repeat(
auto-fit,
minmax(250px,1fr)
);

gap:16px;

}

Metric card:

.metric-card {

padding:15px;

border:
1px solid var(--border);

border-radius:
12px;

background:
var(--surface);
}

Knowledge Graph

.knowledge-node {

display:inline-block;

padding:
8px 14px;

background:
var(--primary);

color:white;

border-radius:
999px;

margin:

5px;
}

Prerequisite arrow:

.knowledge-arrow {

margin:
0 10px;

color:
var(--text-muted);
}

Learning Path Cards

.path-grid {

display:grid;

grid-template-columns:
repeat(
auto-fit,
minmax(280px,1fr)
);

gap:20px;
}

Responsive Layout

Tablet:

@media (max-width:900px)

```
{
.app
{
grid-template-columns:
1fr;
}

.sidebar
{
position:relative;

height:auto;

border-right:none;

border-bottom:
1px solid var(--border);
}
}
```

Phone:

```
@media (max-width:600px)
{
.main
{
padding:15px;
}

.formula
{
font-size:1rem;
}

.button
{
width:100%;
margin-bottom:8px;
```

}
}

Recommended Layout Structure

<body>

<aside class="sidebar">
Learning Paths
Search
Glossary Index
Metrics
Geometry
Linear Algebra
UMAP
Hyperbolic Geometry
</aside>

<main class="main">

Search Box

Formula Walkthrough

Calculator

Quiz

Knowledge Graph

Related Topics

</main>

</body>

This gives your glossary the structure of a professional interactive mathematics learning platform while remaining fully compatible with the reusable walkthrough component, MathJax formulas, calculators, quizzes, metric explorer, and future 3D visualization modules.

Yes. The generated HTML shell has a few malformed tags that would prevent it from loading correctly. Here's the corrected structure plus complete `styles.css` and `app.js` scaffolding.

Fix 1: MathJax Script Tag

Replace:

<script>
window.MathJax = {
tex: {
inlineMath: [['', '']],
displayMath: [['
', '
']]
}
};
</script>

https://cdn.jsdelivr.net/npm/mathjax@3/es5/tex-mml-chtml.js</script>

With:

<script>
window.MathJax = {
tex: {
inlineMath: [['', '']],
displayMath: [['

', '

']]

}

};

</script>

<script

src="https://cdn.jsdelivr.net/npm/mathjax@3/es5/tex-mml-chtml.js">

</script>

Fix 2: CSS Link

Replace:

styles.css

With:

<link

rel="stylesheet"

href="styles.css">

Fix 3: JS File

Replace:

app.jsscript>

With:

<script src="app.js"></script>

Fix 4: Sidebar Links

Several anchors are missing opening tags.

Replace:

#path-beginner
Beginner

With:

Beginner

Do the same for all navigation items.

Final <head>

<head>

<meta charset="UTF-8">

<meta
name="viewport"
content="width=device-width, initial-scale=1.0">

<title>
Interactive Mathematics Learning Kit
</title>

<script>
window.MathJax = {
tex: {
inlineMath: [['', '']],
displayMath: [['

', '

']]
}
};
</script>

<script
src="https://cdn.jsdelivr.net/npm/mathjax@3/es5/tex-mml-chtml.js">
</script>

<link
rel="stylesheet"
href="styles.css">

</head>

---

styles.css

Start with:

:root {

--bg:#f5f7fb;
--surface:#ffffff;
--surface2:#eef4ff;

--text:#1b1b1b;
--muted:#666;

--primary:#3366ff;
--primaryHover:#2148c0;

--success:#0a8f47;
--warning:#c98b00;
--danger:#d11f1f;

--border:#d6dce5;

--radius:14px;

--shadow:
0 4px 12px
rgba(0,0,0,.08);
}

Dark Theme

[data-theme="dark"] {

--bg:#111827;
--surface:#1f2937;
--surface2:#293549;

--text:#f4f4f4;
--muted:#c8c8c8;

--border:#344255;
}

Layout

body {

margin:0;

background:var(--bg);

color:var(--text);

font-family:
"Segoe UI",
sans-serif;
}

```
.app {

display:grid;

grid-template-columns:
280px 1fr;

min-height:100vh;
}
```

Sidebar

```
.sidebar {

background:var(--surface);

border-right:
1px solid var(--border);

padding:20px;

position:sticky;

top:0;

height:100vh;

overflow:auto;
}
.sidebar a {

display:block;

color:var(--text);

text-decoration:none;
```

padding:10px;

border-radius:8px;
}
``
.sidebar a:hover {

background:
var(--surface2);
}

Main

.main {

padding:30px;

max-width:1200px;
}

Card

.card {

background:var(--surface);

border:
1px solid var(--border);

border-radius:
var(--radius);

box-shadow:
var(--shadow);

padding:20px;

```
margin-bottom:20px;
}
```

Buttons

```
.button {

background:
var(--primary);

color:white;

border:none;

padding:
12px 18px;

border-radius:
10px;

cursor:pointer;
}
```

```
.button:hover {

background:
var(--primaryHover);
}
```

Search

```
.search-box {

width:100%;
```

```
padding:12px;

border:
1px solid var(--border);

border-radius:10px;
}
```

Formula

```
.formula {

padding:15px;

margin:15px 0;

background:
var(--surface2);

border-radius:
10px;

overflow-x:auto;

text-align:center;
}
```

Visual Blocks

```
.visual {

background:#111;

color:#00ff95;
```

```
border-radius:10px;

padding:15px;

white-space:pre;

overflow:auto;

font-family:
Consolas,
monospace;
}
```

Metric Grid

```
.metric-grid {

display:grid;

grid-template-columns:
repeat(
auto-fit,
minmax(250px,1fr)
);

gap:20px;
}
`
```

Responsive

```
@media(max-width:900px)
{
.app
{
grid-template-columns:
```

```
1fr;
}

.sidebar
{
height:auto;
position:relative;
}
}
@media(max-width:600px)
{
.main
{
padding:15px;
}

.button
{
width:100%;
margin-bottom:10px;
}
}
```

---

app.js

```
function toggleTheme()
{
const body =
document.body;

body.dataset.theme =
body.dataset.theme === "dark"
? "light"
: "dark";
}
```

Search

```
function filterGlossary()
{
const query =
document
.getElementById(
"searchBox"
)
.value
.toLowerCase();

document
.querySelectorAll(
".glossary-entry"
)
.forEach(entry =>
{
const text =
entry.innerText
.toLowerCase();

entry.style.display =
text.includes(query)
? "block"
: "none";
});
}
`
```

Formula Loader Stub

```
function loadSelectedFormula()
{
const value =
```

```
document
.getElementById(
"formulaSelector"
)
.value;

document
.getElementById(
"walkthroughContainer"
)
.innerHTML =
</span><br></div><div><span
attribution="{&quot;name&quot;:&quot;Copilot&quot;,&quot;oid&quot;:&quot;E
64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E
12-4514-AD9B-893A6FAFD00C&quot;,&quot;userInfo&quot;:{&quot;name&quot;:&qu
ot;Copilot&quot;,&quot;oid&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00
C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;},&
quot;timestamp&quot;:1793397900000,&quot;dataSource&quot;:0}"> &lt;div
class="card"&gt;</span><br></div><div><span
attribution="{&quot;name&quot;:&quot;Copilot&quot;,&quot;oid&quot;:&quot;E
64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E
12-4514-AD9B-893A6FAFD00C&quot;,&quot;userInfo&quot;:{&quot;name&quot;:&qu
ot;Copilot&quot;,&quot;oid&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00
C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;},&
quot;timestamp&quot;:1793397900000,&quot;dataSource&quot;:0}">
&lt;h3&gt;${value}&lt;/h3&gt;</span><br></div><div><span
attribution="{&quot;name&quot;:&quot;Copilot&quot;,&quot;oid&quot;:&quot;E
64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E
12-4514-AD9B-893A6FAFD00C&quot;,&quot;userInfo&quot;:{&quot;name&quot;:&qu
ot;Copilot&quot;,&quot;oid&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00
C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;},&
quot;timestamp&quot;:1793397900000,&quot;dataSource&quot;:0}"></span><br><
/div><div><span
attribution="{&quot;name&quot;:&quot;Copilot&quot;,&quot;oid&quot;:&quot;E
```

64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397900000,"dataSource":0}">
<p>
 <span
attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397900000,"dataSource":0}"> Walkthrough coming from the
 <span
attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397900000,"dataSource":0}">
FormulaWalkthrough component.
 <span
attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397900000,"dataSource":0}">
</p>
 <span
attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397900000,"dataSource":0}">

</div>
 <span attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397900000,"dataSource":0}"> ;

```
if(window.MathJax)
{
MathJax.typesetPromise();
}
}
```

Safe Vector Parser

```
function parseVector(text)
{
return new Float64Array(

text
.trim()
.split(/[\s,]+/)
.filter(x => x !== "")
.map(Number)

);
}
```

Calculator (optimized entry point)

```
function calculateMetrics()
{
const A =
parseVector(
```

```
document
.getElementById(
"vectorA"
).value
);

const B =
parseVector(
document
.getElementById(
"vectorB"
).value
);

if(A.length !== B.length)
{
document
.getElementById(
"results"
)
.innerHTML =
</span><br></div><div><span
attribution="{&quot;name&quot;:&quot;Copilot&quot;,&quot;oid&quot;:&quot;E
64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E
12-4514-AD9B-893A6FAFD00C&quot;,&quot;userInfo&quot;:{&quot;name&quot;:&qu
ot;Copilot&quot;,&quot;oid&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00
C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;},&
quot;timestamp&quot;:1793397900000,&quot;dataSource&quot;:0}"> &lt;div
class="card"&gt;</span><br></div><div><span
attribution="{&quot;name&quot;:&quot;Copilot&quot;,&quot;oid&quot;:&quot;E
64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E
12-4514-AD9B-893A6FAFD00C&quot;,&quot;userInfo&quot;:{&quot;name&quot;:&qu
ot;Copilot&quot;,&quot;oid&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00
C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;},&
quot;timestamp&quot;:1793397900000,&quot;dataSource&quot;:0}"> Dimension
```

```
mismatch.</span><br></div><div><span
attribution="{&quot;name&quot;:&quot;Copilot&quot;,&quot;oid&quot;:&quot;E
64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E
12-4514-AD9B-893A6FAFD00C&quot;,&quot;userInfo&quot;:{&quot;name&quot;:&qu
ot;Copilot&quot;,&quot;oid&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00
C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;},&
quot;timestamp&quot;:1793397900000,&quot;dataSource&quot;:0}">
&lt;/div&gt;</span><br></div><div><span
attribution="{&quot;name&quot;:&quot;Copilot&quot;,&quot;oid&quot;:&quot;E
64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E
12-4514-AD9B-893A6FAFD00C&quot;,&quot;userInfo&quot;:{&quot;name&quot;:&qu
ot;Copilot&quot;,&quot;oid&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00
C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;},&
quot;timestamp&quot;:1793397900000,&quot;dataSource&quot;:0}"> ;
return;
}

let dot = 0;
let normA2 = 0;
let normB2 = 0;
let euclid2 = 0;

for(let i=0;i<A.length;i++)
{
const a = A[i];
const b = B[i];

const d = a-b;

dot += ab;

normA2 += aa;
normB2 += bb;

euclid2 += dd;
```

```
}

const normA =
Math.sqrt(normA2);

const normB =
Math.sqrt(normB2);

let cosine =
"Undefined";

if(
normA > 1e-12 &&
normB > 1e-12
)
{
cosine =
(
dot/
(normA*normB)
).toFixed(8);
}

document
.getElementById(
"results"
)
.innerHTML =
</span><br></div><div><span
attribution="{&quot;name&quot;:&quot;Copilot&quot;,&quot;oid&quot;:&quot;E
64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E
12-4514-AD9B-893A6FAFD00C&quot;,&quot;userInfo&quot;:{&quot;name&quot;:&qu
ot;Copilot&quot;,&quot;oid&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00
C&quot;,&quot;id&quot;:&quot;E64C3D4F-5E12-4514-AD9B-893A6FAFD00C&quot;},&
quot;timestamp&quot;:1793397900000,&quot;dataSource&quot;:0}"> &lt;div
class="result-box"&gt;</span><br></div><div><span
```

attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397900000,"dataSource":0}">
attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397900000,"dataSource":0}">
<p>
 attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397900000,"dataSource":0}">
Dimension:
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${A.length}
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Similarity:
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}

This gives you a valid, runnable three-file application:

index.html
styles.css
app.js

with proper MathJax loading, navigation, responsive design, theme switching, glossary search, formula loading hooks, and an optimized vector calculator ready for the walkthrough framework.

To transform the glossary into a true **Formula Explorer**, every glossary entry should follow the same teaching pattern:

1. **Plain-English Meaning**
2. **Why It Matters**
3. **Visual Intuition**
4. **Formula Representation**
5. **Symbol Breakdown**
6. **Interactive Example**
7. **Common Mistakes**
8. **Related Concepts**
9. **Prerequisites**
10. **Where It Appears Later**

This turns the glossary into a connected curriculum rather than a dictionary.

Example Entry Structure

{
id: "vector",

title: "Vector",

beginner:
"A list of numbers representing a direction and magnitude.",

technical:

"An element of a vector space.",

visual:
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→
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why:

"Vectors are the foundation of geometry,
machine learning, embeddings and UMAP.",

formula:
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 <span attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00

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[x_1,x_2,...,x_n]
 <span attribution="{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","userInfo":{"name":"Copilot","oid":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C","id":"E64C3D4F-5E12-4514-AD9B-893A6FAFD00C"},"timestamp":1793397900000,"dataSource":0}"> ,

symbols: [
{
symbol:"v",
meaning:"vector"
},
{
symbol:"x_i",
meaning:"first coordinate"
}
],

examples:[

{
title:"2D vector",
expression:"[3,4]"
},

{
title:"768D embedding",
expression:"[0.42,...]"
}
],

prerequisites:[
"Coordinate",

"Dimension"
],

unlocks:[
"Norm",
"Distance",
"Dot Product",
"Cosine Similarity"
]
}

New Formula Explorer Categories

Category 1: Mathematical Symbols

These are usually skipped, but beginners need them.

Equal Sign

=

Meaning

"Has the same value as."

Example

2+3=5

Common Mistake

Thinking it means "calculate now."

Plus

+

Adds quantities.

Minus

-

Subtracts quantities.

Also appears as:

-5

meaning a negative number.

Multiplication

\times
\cdot
ab

All can mean multiplication.

Division

\frac{a}{b}

Meaning

a divided by b.

Interactive

Show numerator and denominator visually.

Power

x^2

Meaning

x \times x

Root

\sqrt{x}

Meaning

The number that multiplied by itself gives x.

Absolute Value

|x|

Meaning

Distance from zero.

Example:

|-5|=5
``

Summation

\sum

One of the most important symbols.

Expand:

\sum_{i=1}^{3} i

into:

1 + 2 + 3

with animation.

Infinity

\infty

Interactive explanation:

Not a number.

Represents something without bound.

Category 2: Core Geometry Terms

Point

Formula view:

P=(x,y)

Symbol walkthrough:

P = point

x = horizontal position

y = vertical position

Coordinate

Interactive:

(2,3)

Highlight each coordinate separately.

Dimension

Formula:

(x_1,x_2,\ldots,x_n)

Walkthrough:

1D → Line

2D → Plane

3D → Space

nD → Generalized space

Category 3: Linear Algebra

Norm

Formula:

||x||
=
\sqrt{
x_1^2+x_2^2+\cdots+x_n^2
}

Formula explorer:

Step 1

Square coordinates.

Step 2

Add them.

Step 3

Take square root.

Meaning

Vector length.

Dot Product

Formula:

x\cdot y
=
\sum x_i y_i

Interactive expansion:

[1,2]
.
[3,4]

=
1×3 + 2×4

=
11

Visual meaning explorer:

Positive
Zero
Negative
`

Category 4: Metrics

Every metric should have:

What question does it answer?

Formula

Visual intuition

Animation

Typical use cases

Euclidean Distance

Question:

> How far apart are two points?

Formula:

$$d=\sqrt{\sum(x_i-y_i)^2}$$

Interactive reveal of every operation.

Manhattan Distance

Question:

> How many blocks must I walk?

Formula:

$$d=$$

\sum |x_i-y_i|

Visual:

Grid city.

Chebyshev Distance

Question:

> What is the largest coordinate difference?

Formula:

\max_i |x_i-y_i|

Visual:

Chess king movement.

Minkowski Distance

Question:

> What family of distance measures contains Euclidean and Manhattan?

Formula:

\left(
\sum
|x_i-y_i|^p
\right)^{1/p}

Interactive slider:

p = 1

p = 2

p = 10

p = 100

Cosine Similarity

Question:

> Are these vectors pointing in the same direction?

Formula:

$$
\frac{
x\cdot y
}
{
||x|| ||y||
}
$$

Interactive angle wheel.

Jaccard Similarity

Question:

> How much overlap exists?

Formula:

$$
\frac{
|A\cap B|
}{
|A\cup B|
}
$$

Interactive set explorer.

Category 5: Statistics

Mean

Formula:

\mu
=
\frac{
\sum x_i
}
{n}

Step-by-step breakdown.

Variance

Formula:

\sigma^2
=
\frac{
\sum
(x_i-\mu)^2
}
{n}

Interactive deviation animation.

Standard Deviation

Formula:

\sigma
=
\sqrt{\sigma^2}

Visualize spread.

Covariance

Formula explorer should show:

Increase together

Decrease together

Move opposite

Category 6: Manifolds & Geometry

Manifold

Technical:

Locally Euclidean space.

Interactive:

Earth surface zoom.

Geodesic

Question:

What is the shortest path on a curved space?

Examples:

Sphere
Hyperbolic space
``

Curvature

Formula panel:

K

Meaning:

How space bends.

Interactive:

Flat

Positive

Negative

Category 7: Hyperbolic Geometry

Hyperboloid

Formula explorer:

z^2-x^2-y^2=1

Step-by-step:

1. Compare to sphere.

2. Explain sign change.
3. Show resulting shape.
4. Show rotating 3D hyperboloid.

Poincaré Disk

Formula explorer:

X = x/(1+z)

Y = y/(1+z)

Interactive projection.

Category 8: Embeddings and UMAP

Embedding

Formula view:

Object → Vector

Examples:

Word
Image
Document
Person

Input Metric

Formula explorer asks:

> How do we decide neighbors?

Examples:

Euclidean
Cosine
Jaccard

Output Metric

Question:

> What geometry should contain the embedding?

Examples:

Euclidean
Sphere
Hyperboloid

UMAP

Interactive pipeline:

Data

↓
Neighborhood Graph

↓
Optimization

↓
Embedding

Each stage clickable.

Add a Universal "Explain This Formula" Panel

Every formula card gets:

Question It Answers
What Every Symbol Means
Expand Summation
Show Worked Example
Visual Interpretation
Interactive Calculator
Common Mistakes
Applications
Related Concepts

This gives users a consistent learning experience across symbols, operators, metrics, geometry, linear algebra, hyperbolic geometry, embeddings, and UMAP, making the glossary genuinely self-contained from first principles through advanced topics.

Excellent idea. If this is going to be a **ground-up mathematics → geometry → embeddings → UMAP → hyperbolic spaces learning system**, then every equation should appear in its:

1. **Specific form** (e.g. 2D distance)
2. **Generalized n-dimensional form**
3. **Abstract mathematical form** (where appropriate)
4. **Computational form**
5. **Geometric interpretation**

Below is a review of the mathematics currently present and the generalized forms that should be added.

Foundation Mathematics

Addition

Current:

3+4=73+4=7

Generalized:

∑_i=1nx_i\sum_{i=1}^{n} x_i

Abstract:

a+ba+b

Multiplication

Current:

a×ba\times b

Generalized:

∏_i=1nx_i\prod_{i=1}^{n} x_i

(Product operator)

Exponentiation

Current:

x2x^2

Generalized:

xpx^p

where

p∈Rp\in\mathbb{R}

Root

Current:

x\sqrt{x}

Generalized:

x1/px^{1/p}

Coordinates

Point

Current:

(x,y)(x,y)

Generalized:

(x_1,x_2,…,x_n)(x_1,x_2,\ldots,x_n)

Vector form:

x==========[x_1x_2⋮x_n]\mathbf{x} ========== \begin{bmatrix} x_1\\ x_2\\ \vdots\\ x_n \end{bmatrix}

Distances

Euclidean Distance

Current:

d=(x_2−x_1)2\+(y_2−y_1)2d= \sqrt{ (x_2-x_1)^2 \+ (y_2-y_1)^2 }

Generalized:

d(x,y)=========================∑_i=1n(x_i−y_i)2d(\mathbf{x},\mathbf{y})
========================= \sqrt{ \sum_{i=1}^{n} (x_i-y_i)^2 }

Norm form:

d=|x−y|_2d= |\mathbf{x}-\mathbf{y}| _2

Manhattan Distance

Current:

∑|x_i−y_i|\sum |x_i-y_i|

Generalized:

d=∑_i=1n|x_i−y_i|d= \sum_{i=1}^{n} |x_i-y_i|

Norm form:

|x−y|_1|\mathbf{x}-\mathbf{y}|_1

Chebyshev Distance

Current:

max|x_i−y_i|\max |x_i-y_i|

Generalized:

d=max_i|x_i−y_i|d= \max_i |x_i-y_i|

Norm form:

|x−y|_∞|\mathbf{x}-\mathbf{y}|_\infty

Minkowski Distance

Current:

(∑|x_i−y_i|p)1/p\left(\sum |x_i-y_i|^p \right)^{1/p}

Generalized:

d_p(x,y)==========(∑_i=1n|x_i−y_i|p)1/pd_p(\mathbf{x},\mathbf{y}) ========== \left(\sum_{i=1}^{n} |x_i-y_i|^p \right)^{1/p}

Special cases:

p=1p=1

Manhattan

p=2p=2

Euclidean

p→∞p\to\infty

Chebyshev

Vectors

Vector

Current:

[1,2,3]

Generalized:

x==========(x_1,x_2,…,x_n)\mathbf{x} ========== (x_1,x_2,\ldots,x_n)

Vector Addition

Add this.

x\+y==========(x_1+y_1,…,x_n+y_n)\mathbf{x} \+ \mathbf{y} ========== (x_1+y_1, \ldots, x_n+y_n)

Scalar Multiplication

Add this.

ax===========(ax_1,…,ax_n)a\mathbf{x} =========== (ax_1,\ldots,ax_n)

Norms

Euclidean Norm

Current:

||x||||x||

Generalized:

|x|_2===============∑_i=1nx_i2|\mathbf{x}|_2 =============== \sqrt{ \sum_{i=1}^{n} x_i^2 }

General p-Norm

Missing.

|x|_p===============(∑_i=1n|x_i|p)1/p|\mathbf{x}|_p =============== \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}

Infinity Norm

Missing.

|x|_∞====================max_i|x_i|\mathbf{x}|_\infty ==================== \max_i |x_i|

Dot Product

Current:

x⋅y========∑x_iy_ix\cdot y ======== \sum x_i y_i

Generalized:

xTy======================∑_i=1nx_iy_i\mathbf{x}^T\mathbf{y} ====================== \sum_{i=1}^{n} x_i y_i

Matrix form:

xTy\mathbf{x}^T\mathbf{y}

Angles

Cosine Similarity

Current:

x⋅y||x||||y||\frac{x\cdot y} {||x||||y||}

Generalized:

cos(θ)============xTy|x|_2|y|_2\cos(\theta) ============ \frac{ \mathbf{x}^T\mathbf{y} } { |\mathbf{x}|_2 |\mathbf{y}|_2 }

Angle

Current:

θ=arccos(cosθ)\theta= \arccos(\cos\theta)

Generalized:

θ=arccos(xTy|x||y|)\theta= \arccos \left(\frac{ \mathbf{x}^T\mathbf{y} } { |\mathbf{x}| |\mathbf{y}| } \right)

Matrices

Currently underrepresented.

Matrix

Add:

A=[a_11&a_12a_21&a_22]A= \begin{bmatrix} a_{11}\&a_{12}\\ a_{21}\&a_{22} \end{bmatrix}

General:

A∈Rm×nA\in \mathbb{R}^{m\times n}

Matrix Multiplication

Add:

ABAB

Component form:

(AB)_ij==========∑_kA_ikB_kj(AB)_{ij} ========== \sum_k A_{ik}B_{kj}

Statistics

Mean

Current:

μ=∑x_in\mu= \frac{\sum x_i}{n}

Good.

Variance

Current:

σ2========∑(x_i−μ)2n\sigma^2 ======== \frac{ \sum (x_i-\mu)^2 }{n}

General population form.

Add sample form:

s2===∑(x_i−x¯)2n−1s^2 === \frac{ \sum (x_i-\bar x)^2 }{n-1}

Standard Deviation

σ=σ2\sigma= \sqrt{\sigma^2}

Good.

Covariance

Currently incomplete.

Add:

Cov(X,Y)==========E\[(X−μ_X)(Y−μ_Y)]\mathrm{Cov}(X,Y) ========== E\[(X-\mu_X) (Y-\mu_Y)]

Sample form:

∑_i(x_i−x¯)(y_i−y¯)n−1\frac{ \sum_i (x_i-\bar x) (y_i-\bar y) } {n-1}

Similarity Metrics

Jaccard

Current:

∣A∩B∣∣A∪B∣\frac{|A\cap B|} {|A\cup B|}

Good.

Also add distance:

d_J====1−Jd_J ==== 1-J

Hamming Distance

Generalize:

d_H====∑_i1(x_i≠y_i)d_H ==== \sum_i 1(x_i\neq y_i)

where

1(⋅)1(\cdot)

is the indicator function.

Geometry

Circle

Missing.

x2+y2=r2x^2+y^2=r^2

Sphere

Current intuition only.

Add:

x_12+x_22+⋯+x_n2==========r2x_1^2+x_2^2+\cdots+x_n^2 ========== r^2

Hyperbolic Geometry

Hyperboloid

Current:

z2−x2−y2=1z^2-x^2-y^2=1

Generalized:

x_02−−−−−−∑_i=1nx_i2======1x_0^2 ------ \sum_{i=1}^{n} x_i^2 ====== 1

This is the true n-dimensional hyperboloid model.

Lorentz Inner Product

Missing but critical.

\langle x,y\rangle_L ========== ## x_0y_0 \sum_{i=1}^{n} x_i y_i

Hyperbolic distance depends on this.

Hyperbolic Distance

Missing.

d_H(x,y)==========arcosh(⟨x,y⟩_L)d_H(x,y) ========== \operatorname{arcosh} (\langle x,y\rangle_L)

Absolute must-have for a hyperbolic geometry section.

Poincaré Disk

Current:

X=x1+zX= \frac{x}{1+z} Y=y1+zY= \frac{y}{1+z}

Generalized:

p=xx_0+1p= \frac{\mathbf{x}} {x_0+1}

for hyperboloid point

(x_0,x)(x_0,\mathbf{x})

Embeddings

Embedding Function

Missing.

f:X→Rnf: X \rightarrow \mathbb{R}^n

Meaning:

Object → Vector

Dimensionality Reduction

Projection

General:

f:Rn→Rmf: \mathbb{R}^n \rightarrow \mathbb{R}^m

where

m\<nm\<n

PCA

Add:

Y=XWY = XW

where

WW

contains principal directions.

UMAP

The glossary currently lacks the actual mathematical objective.

At minimum include:

Neighbor Graph

G=(V,E)G=(V,E)

Fuzzy Simplicial Set Concept

μ_ij∈\[0,1]\mu_{ij} \in \[0,1]

Neighborhood strength.

Cross-Entropy Objective

Show conceptually:

UMAP==========argmin(graph mismatch)\text{UMAP} ========== \arg\min (\text{graph mismatch})

Even if the full optimization equation is hidden behind an "advanced" toggle.

Most Important Missing Generalized Equations

If I were prioritizing, the biggest omissions are:

1. General p-norm

|x|_p|\mathbf{x}|_p

2. General Minkowski metric

d_p(x,y)d_p(\mathbf{x},\mathbf{y})

3. General sphere equation

∑x_i2=r2\sum x_i^2=r^2

4. General hyperboloid equation

x_02−∑x_i2=1x_0^2-\sum x_i^2=1

5. Lorentz inner product

⟨x,y⟩_L\langle x,y\rangle_L

6. Hyperbolic distance

arcosh(⟨x,y⟩_L)\operatorname{arcosh} (\langle x,y\rangle_L)

7. Vector space operations

axa\mathbf{x}

and

x+y\mathbf{x}+\mathbf{y}

8. Matrix multiplication

ABAB

9. Covariance

Cov(X,Y)\mathrm{Cov}(X,Y)

10. Embedding function

f:X→Rnf:X\rightarrow\mathbb{R}^n

Those ten additions would make the glossary mathematically coherent from arithmetic all the way through hyperbolic UMAP and modern embedding spaces.